

PRACTICAL 10

AIM : Data Preparation and Transformation using Tableau Prep

What you will learn ?

- a) Connect and load data into Tableau Prep
- b) Clean and profile data
- c) Group and standardize values
- d) Merge mismatched fields
- e) Perform aggregations in Prep
- f) Apply filters and transformations
- g) Create and automate data preparation flows

You need Tableau Prep Builder for this Lesson. You need to download and install. Here are the steps

1. Download

■ Download the Windows installer (.exe) directly from the official Tableau Prep Builder Releases Page (or use the Tableau Prep Free Trial Page).

2. Install

■ Open your Downloads folder and run the .exe file.

■ Accept the license agreement, click Install, and confirm Yes on the Windows prompt.

3. Activate & Launch

■ Open Tableau Prep Builder from the Start Menu.

■ Select Activate with a product key (enter your Tableau Desktop/Student key) or Start a Free Trial.

Dataset: Custom Designed Students Dataset [csv files]

Step 1: Click the Link <https://www.tableau.com/products/prep/download>

File 1: Student_Assessments_Term1.csv

Student_ID,Student_Name,Department,Assessment_Type,Exam_Date,
Subject_Code,Marks_Obtained,Max_Marks,Enrollment_Status
S101,Aarav Sharma ,Computer
Science,Midterm,2025/09/15,CS101,85,100,Active
S101,Aarav Sharma,Comp

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Sci,Final,2025/11/20,CS101,92,100,Active
S102,priya VERMA,Electronics,Midterm,15-09-
2025,EC101,70,100,Active
S102,priya VERMA,ECE,Final,20-11-2025,EC101,78,100,Active
S103,Rohan Gupta,CS,Midterm,2025.09.15,CS101,45,100,Active
S103,Rohan Gupta,CSE,Final,2025.11.20,CS101,52,100,Active
S104,Sneha Patel,Information
Tech,Midterm,2025/09/15,IT101,90,100,Active
S104,Sneha Patel,IT,Final,2025/11/20,IT101,96,100,Active
S105,aditya Rao
,Mechanical,Midterm,2025/09/15,ME101,35,100,Dropped
S105,aditya Rao ,Mech
Engg,Final,2025/11/20,ME101,40,100,Dropped
S106,Ananya Singh,Computer
Science,Midterm,2025/09/15,CS101,88,100,Active
S106,Ananya Singh,CS,Final,2025/11/20,CS101,94,100,Active
S107,Rahul Nair,ECE,Midterm,2025/09/15,EC101,65,100,Active
S107,Rahul Nair,Electronics
Engg,Final,2025/11/20,EC101,72,100,Active
S108,Kavya Reddy,IT,Midterm,2025/09/15,IT101,92,100,Active
S108,Kavya Reddy,Info
Tech,Final,2025/11/20,IT101,98,100,Active

File 2: Student_Assessments_Term2.csv

Std_ID,Full_Name,Dept,Exam_Type,Assessment_Date,Course_Code,S
core>Total_Possible,Enrollment_Status

S101,Aarav Sharma,Computer
Science,Midterm,2026/02/10,CS201,88,100,Active
S101,Aarav Sharma,CS,Final,2026/04/25,CS201,95,100,Active
S102,Priya Verma,ECE,Midterm,2026/02/10,EC201,75,100,Active

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S102,Priya

Verma,Electronics,Final,2026/04/25,EC201,82,100,Active

S103,Rohan Gupta,CSE,Midterm,2026/02/10,CS201,50,100,Active

S103,Rohan Gupta,CS,Final,2026/04/25,CS201,60,100,Active

S104,Sneha Patel,IT,Midterm,2026/02/10,IT201,94,100,Active

S104,Sneha Patel,Information

Tech,Final,2026/04/25,IT201,98,100,Active

S106,Ananya Singh,CSE,Midterm,2026/02/10,CS201,90,100,Active

S106,Ananya Singh,Computer

Science,Final,2026/04/25,CS201,96,100,Active

S107,Rahul Nair,ECE,Midterm,2026/02/10,EC201,70,100,Active

S107,Rahul

Nair,Electronics,Final,2026/04/25,EC201,75,100,Active

S108,Kavya Reddy,IT,Midterm,2026/02/10,IT201,95,100,Active

S108,Kavya Reddy,IT,Final,2026/04/25,IT201,100,100,Active

File 3: Course_Credit_Master.csv

Subject_Code,Subject_Title,Credit_Hours,Passing_Marks

CS101,Data Structures,4,40

CS201,Algorithms,4,40

EC101,Digital Logic,3,40

EC201,Signals & Systems,4,40

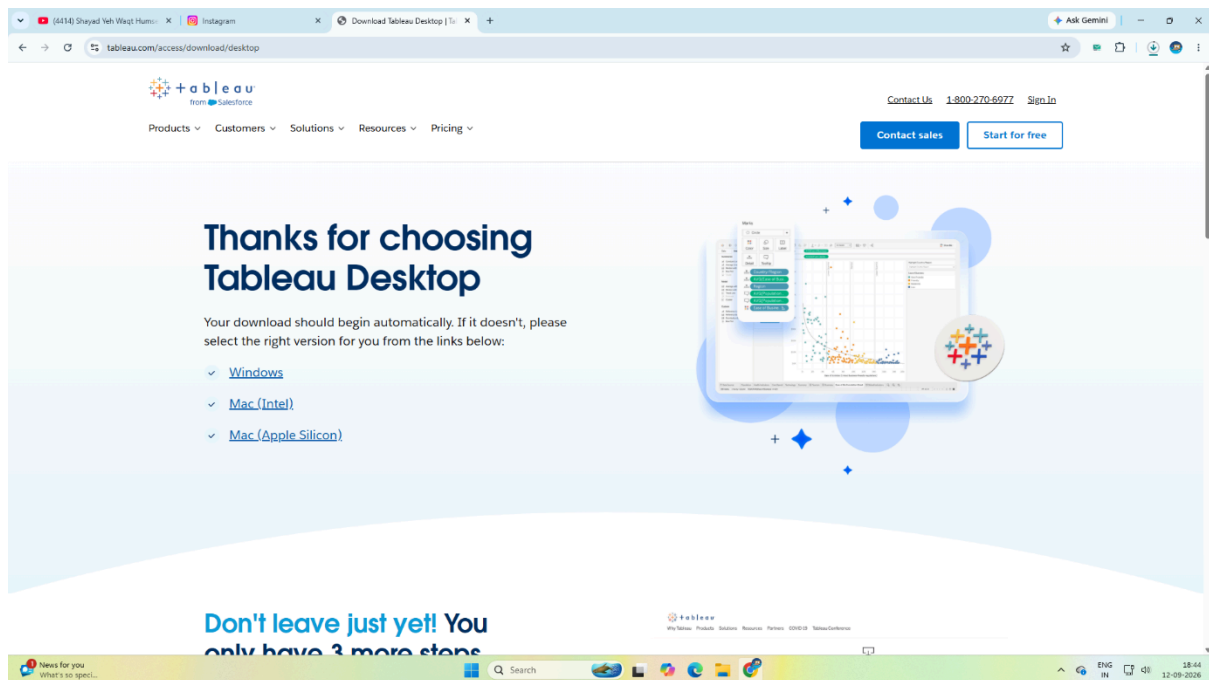
IT101,Database Systems,4,40

IT201,Cloud Computing,4,40

ME101,Thermodynamics,3,40

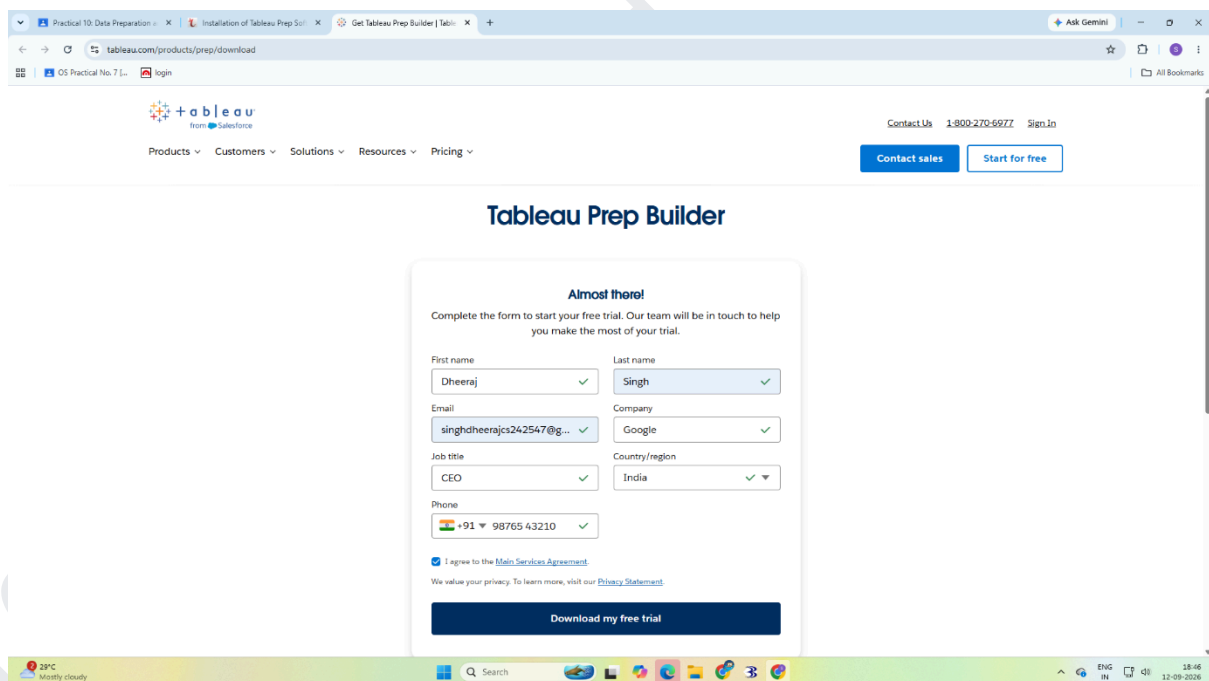
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A: Connect and Load Data into Tableau



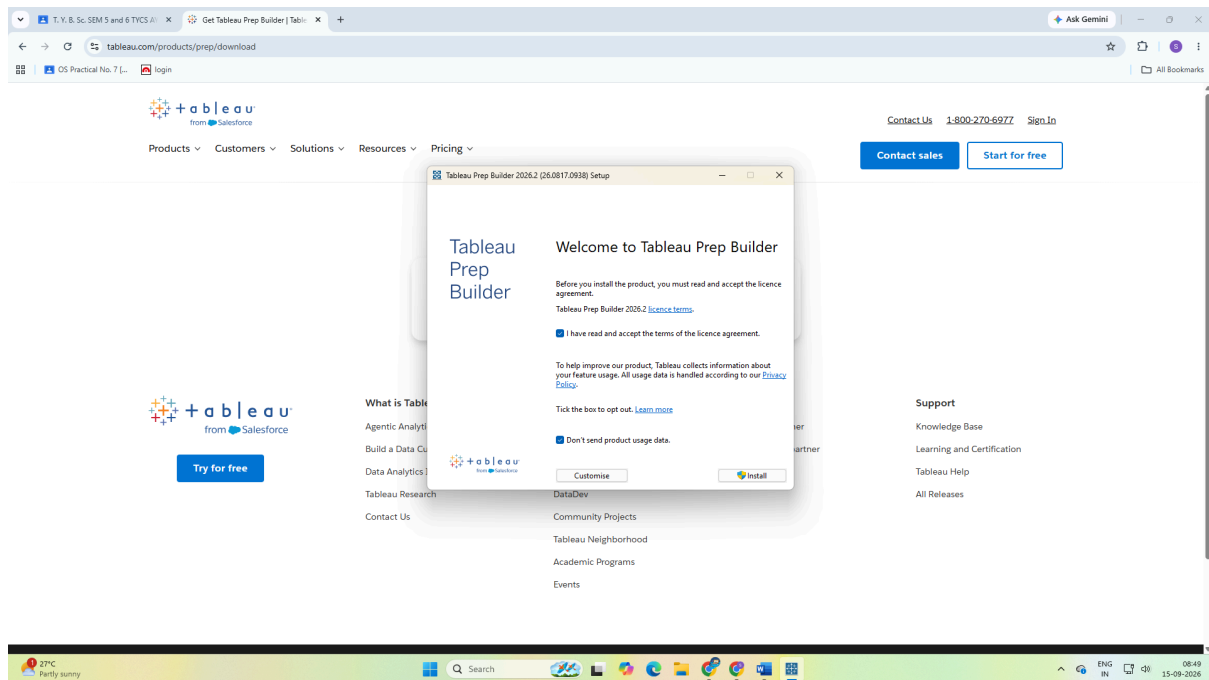
You Will be Directed to this Page

Step 2: Fill the details

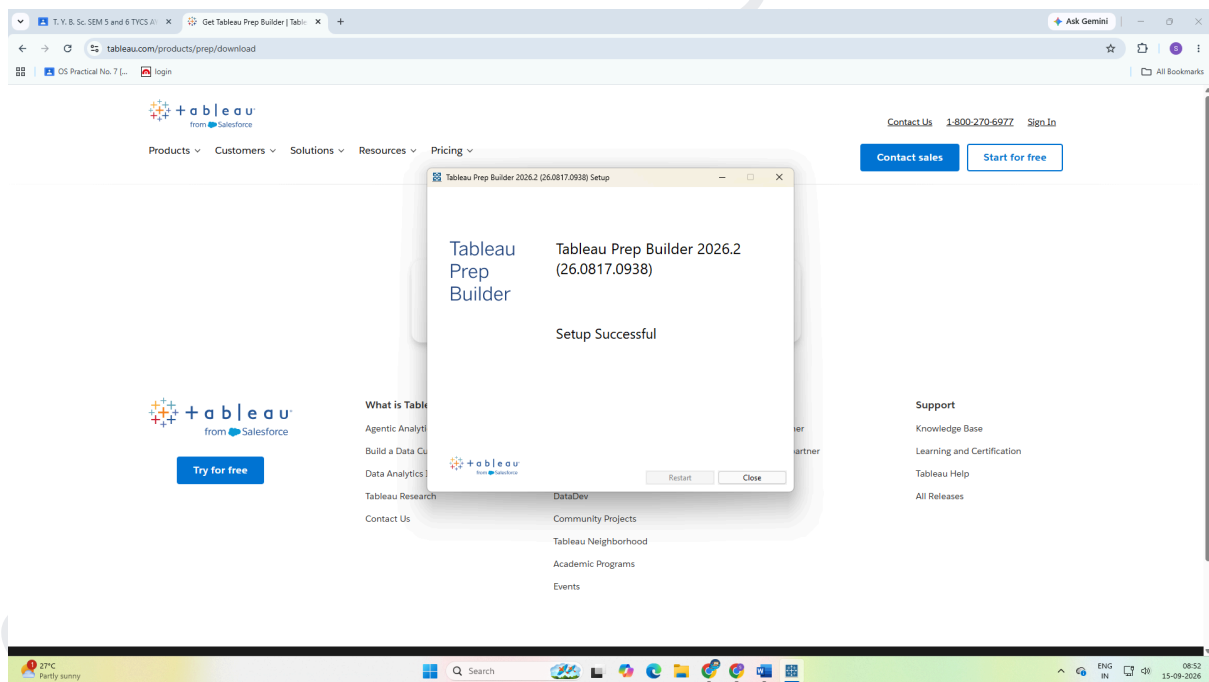


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Step 3: You will be directed to this particular page, Click on the options valid to you or the Download has already started after entering this page (I am doing Windows since I have a Windows Machine.)

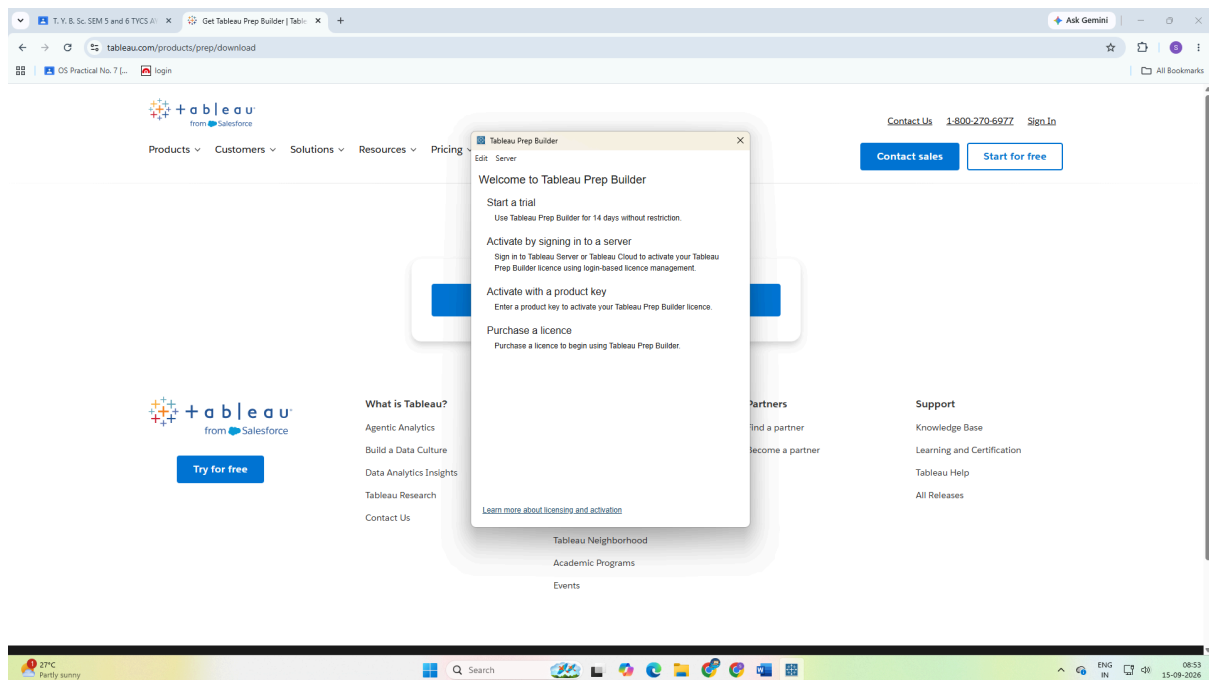


Step 4: After installation is completed and click close.



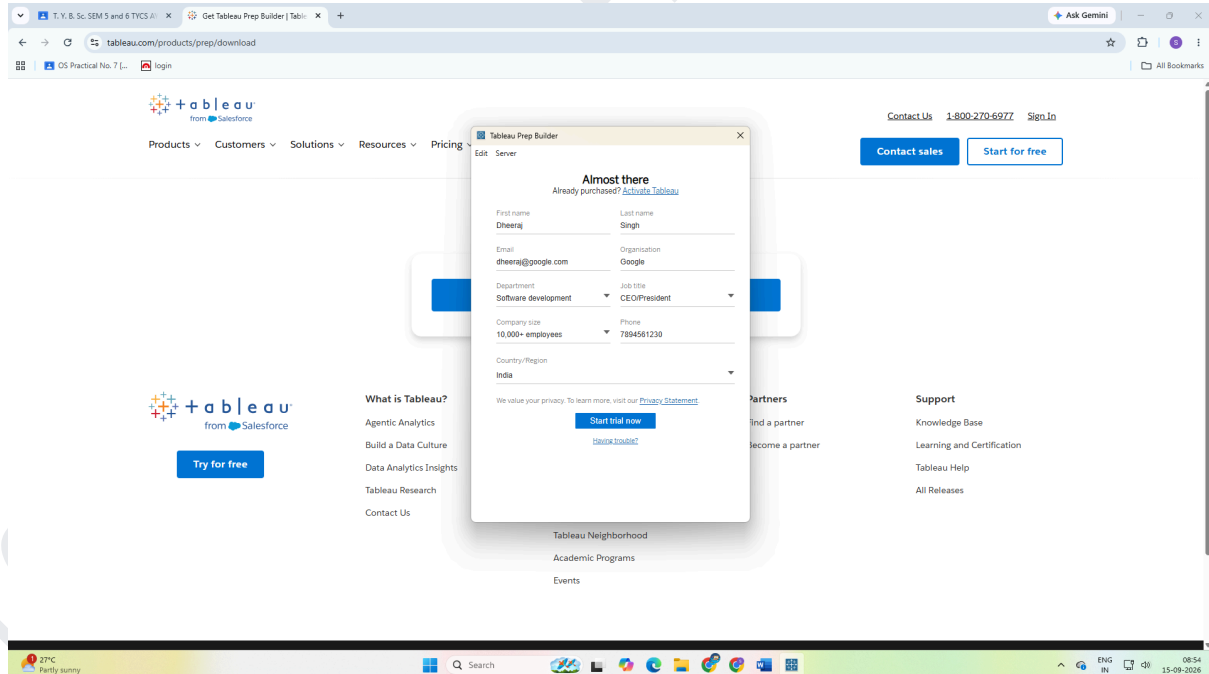
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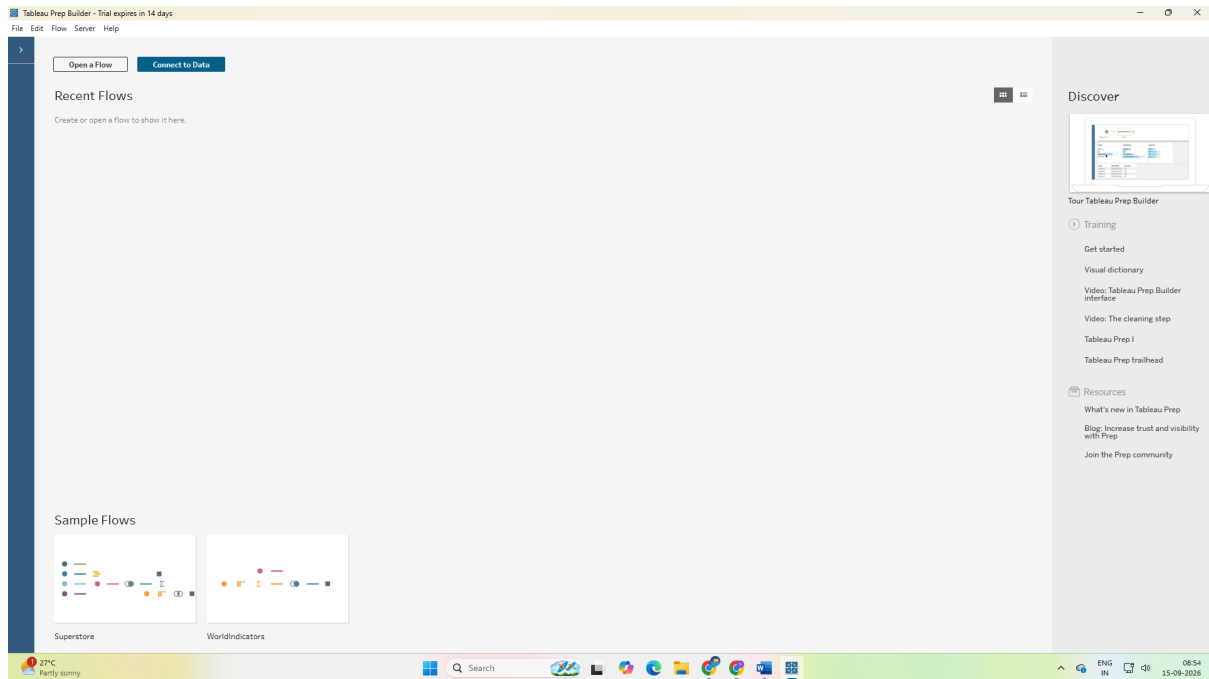
Step 5: Launch the Tableau Prep Builder and click on start a trial



After Clicking in start trial fill the details and in Organization field just enter a dot and fill the other details as shown in snapshot and click start trial now.

After Filling your details click on start trial





We have successfully Installed the Tableau Prep Software.

Objective: Establish connections to the local CSV data sources (Student_Assessments_Term1.csv, Student_Assessments_Term2.csv, and Course_Credit_Master.csv) and initiate a visual data preparation workflow.

Tableau Prep Builder utilizes an input step pipeline architecture. Connecting multiple disparate files into a unified workspace establishes the foundation for schema validation, data type detection, and downstream data profiling.

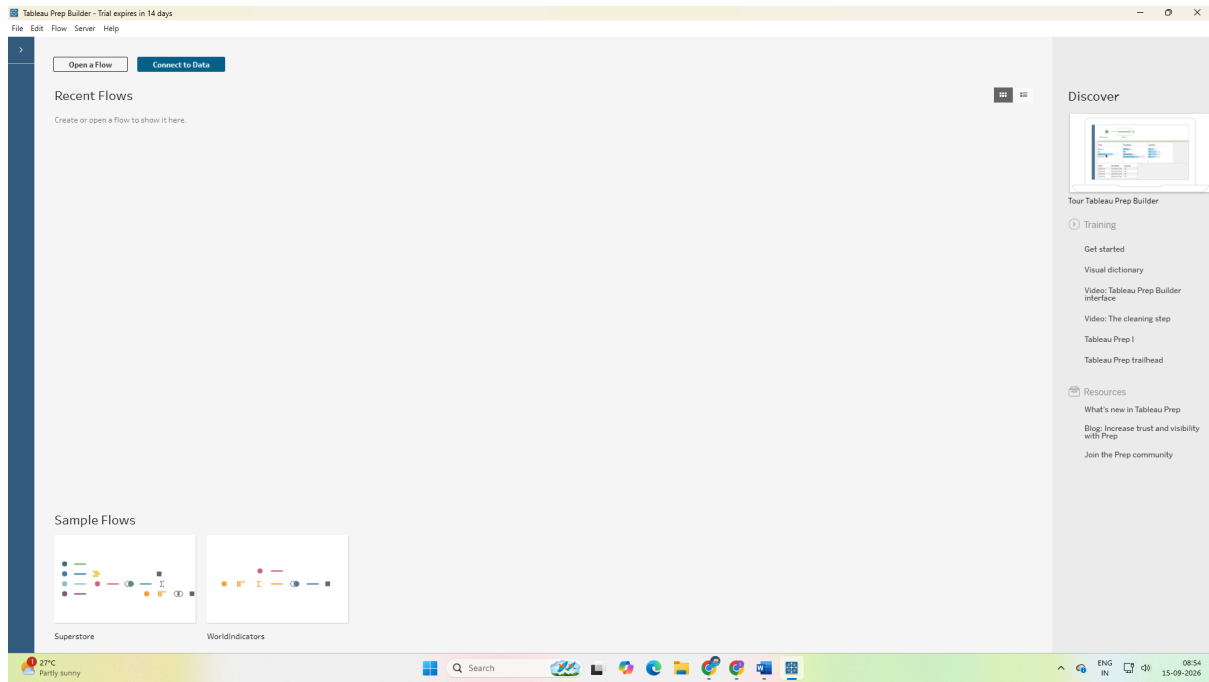
Step 1: Open Tableau Prep Builder

1. Launch Tableau Prep Builder on your machine.
2. On the start screen, locate the Connections pane on the left-hand side.

Step 2: Load the First Dataset (Student_Assessments_Term1.csv)

1. In the Connections pane, click the + (Add connection) button.

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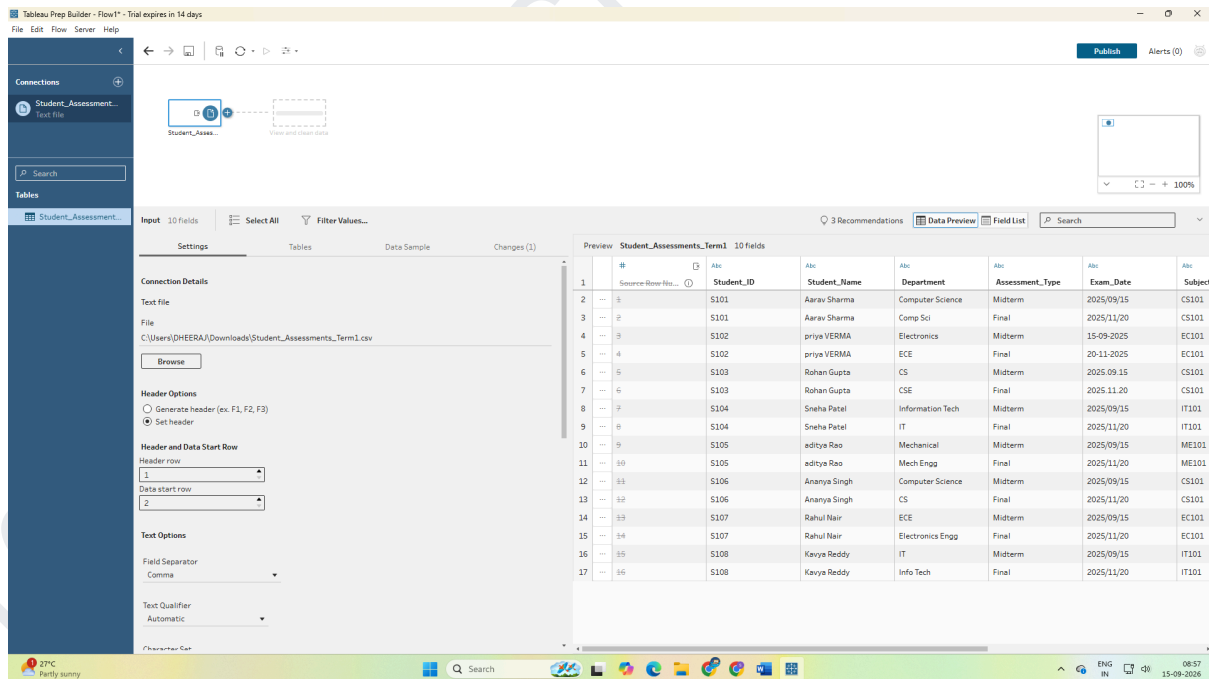


2. In the list of file formats, select Text file.

3. Browse to the folder where you saved the CSVs, select

Student_Assessments_Term1.csv, and click Open.

4. An Input Step icon named Student_Assessments_Term1 will appear on the Flow canvas.



5. In the bottom Input configuration pane, verify:

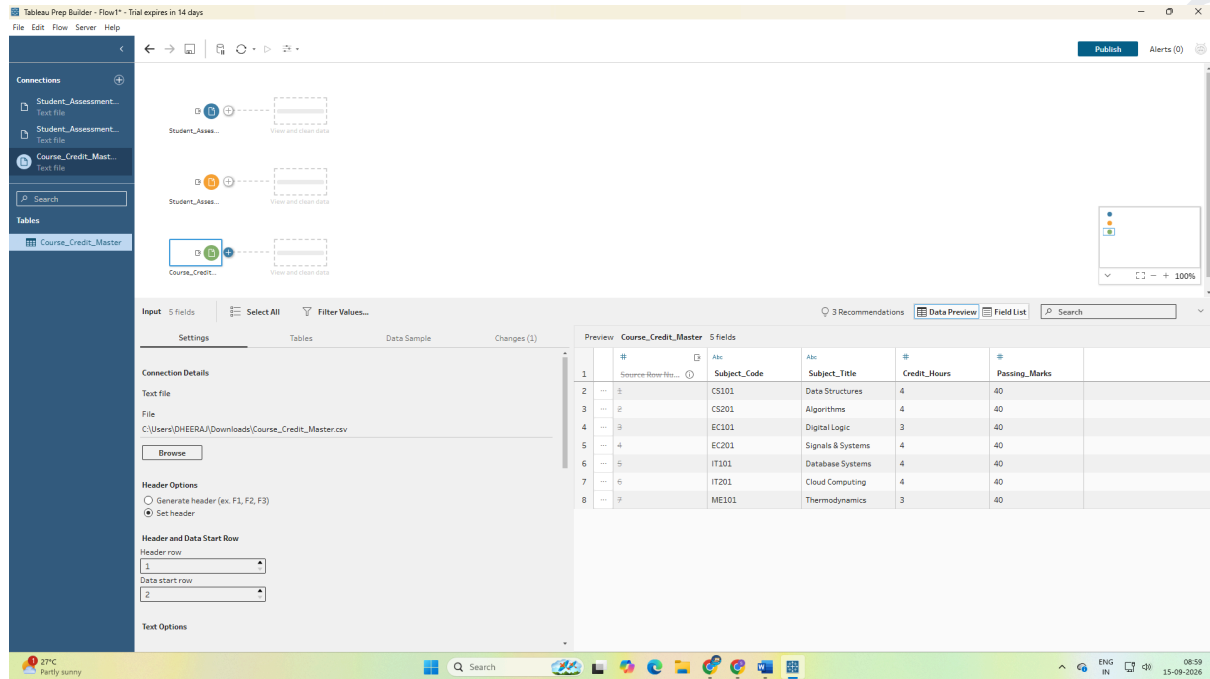
o Field Separator: Comma (,)

o First row contains header: Checked

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Step 3: Load the Second and Third Datasets

1. In the top-left Connections pane, click the + (Add connection) button again.
2. Select Text file, browse and select Student_Assessments_Term2.csv, and click Open.
3. Click + (Add connection) once more  select Text file  choose Course_Credit_Master.csv click Open.

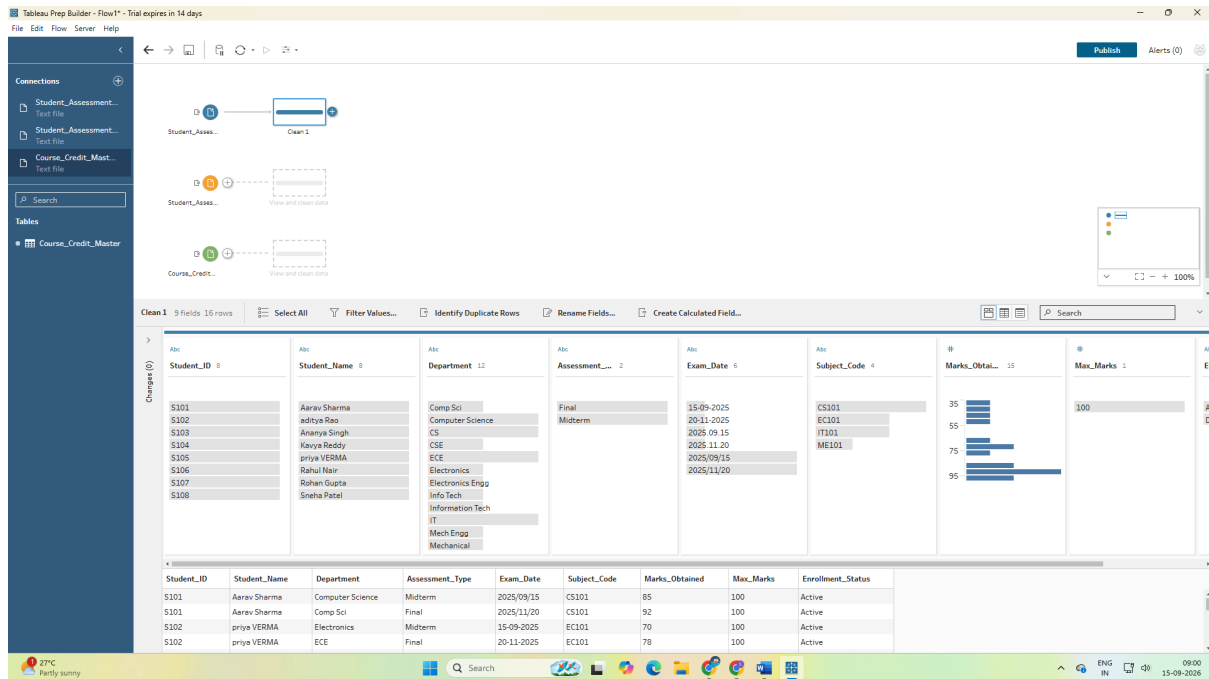


4. You will now see three separate Input icons on your Flow canvas.

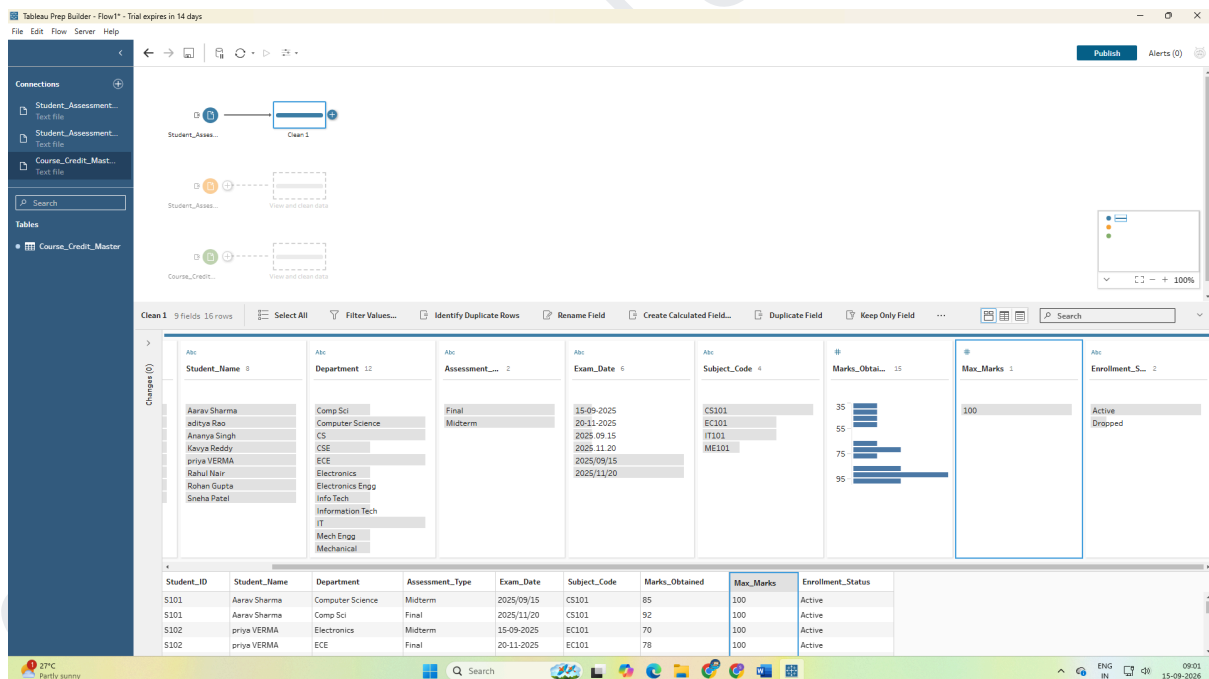
Step 4: Add an Initial Clean Step to Term 1

1. On the Flow canvas, hover your mouse over the Student_Assessments_Term1 input icon.
2. Click the small + icon that appears to its right and Select Clean Step

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4. The Profile Pane will open at the bottom, displaying interactive profile cards for each of the 9 columns in Term 1.



B: Clean and Profile Data

Objective: Use Tableau Prep Builder's interactive Profile Pane to identify data anomalies, whitespace issues, date formatting discrepancies, and inconsistent casing, then apply standard cleaning operations. Unlike traditional spreadsheets, Tableau Prep provides real-time profiling cards that visualize distributions, frequency counts, and metadata data types for every field. Profiling allows you

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to quickly detect hidden formatting defects (such as extra spaces in names like ' Aarav Sharma ' or non-standard date strings) before standardizing categorical groups or performing joins.

Step 1: Inspect the Profile Pane

1. Click on the Clean 1 step connected to Student_Assessments_Term1.

2. Look at the cards displayed in the Profile Pane:

o Each card represents one column.

o Notice that Student_Name has multiple representations of the same student due to extra spaces and mixed casing (e.g., ' Aarav Sharma ' vs 'Aarav Sharma', 'priya VERMA', 'aditya Rao ').

o Notice that Exam_Date is recognized as a string (Abc) due to mixed separators (/ , - , .).

Step 2: Clean Whitespace and Standardize Letter Casing

1. On the Student_Name profile card:

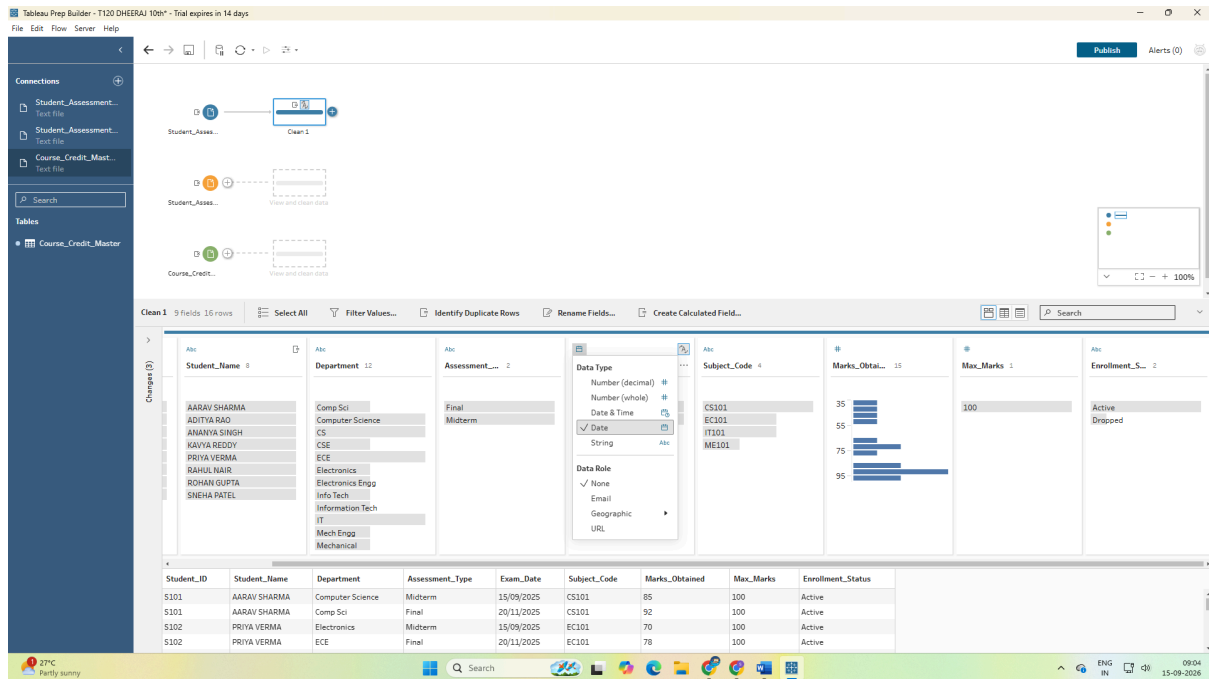
o Click the More options (...) icon at the top right of the card.

o Go to Clean --> select Trim Spaces (this strips all leading and trailing whitespace).

The screenshot shows the Tableau Prep Builder interface. On the left, the 'Connections' pane lists 'Student_Assessment...', 'Student_Assessment...', and 'Course_Credit_Mast...'. The 'Tables' pane shows 'Course_Credit_Master'. The main workspace displays a data flow: 'Student_Assessment...' (Table) -> 'Clean 1' (Step) -> 'Student_Assessment...' (Table). Below this, a 'View and clean data' button is visible. The 'Clean 1' step is selected, showing a list of fields: 'Student_Name', 'Department', 'Assessment_Type', 'Exam_Date', 'Subject_Code', 'Marks_Obtained', 'Max_Marks', and 'Enrollment_Status'. The 'Student_Name' field is highlighted with a red box. The 'Clean 1' step is configured to 'Trim Spaces' for the 'Student_Name' field. The resulting data table is shown at the bottom:

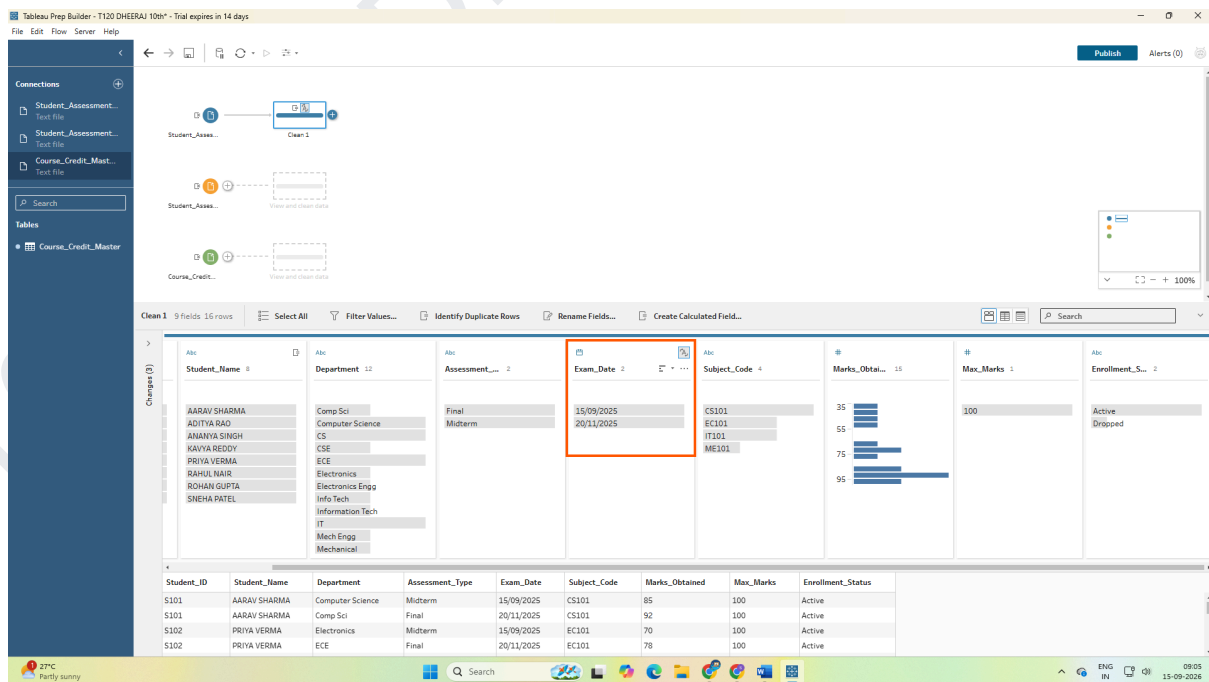
Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S101	AARAV SHARMA	Computer Science	Midterm	2025/09/15	CS101	95	100	Active
S101	AARAV SHARMA	Comp Sci	Final	2025/11/20	CS101	92	100	Active
S102	PRIYA VERMA	Electronics	Midterm	15-09-2025	EC101	70	100	Active
S102	PRIYA VERMA	ECE	Final	20-11-2025	EC101	78	100	Active

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Step 3: Convert Exam_Date to Date Data Type

1. Locate the Exam_Date profile card.
2. Click the data type icon (currently showing Abc) directly above the column name.
3. Select Date from the dropdown menu.
4. Tableau Prep automatically parses the various date formats (2025/09/15, 15-09-2025, 2025.11.20) into standardized calendar dates.



Step 4: Clean the Second Dataset (Student_Assessments_Term2.csv)

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1. On the Flow canvas, hover over the Student_Assessments_Term2 input icon.
2. Click + --> select Clean Step (creates Clean 2).
3. On the Full_Name card in Clean 2: o Click More options (...) --> Clean --> Trim Spaces.
4. On the Assessment_Date card: o Click the Abc icon --> change the data type to Date.

The screenshot shows the Tableau Prep Builder interface. On the left, the 'Connections' pane lists 'Student_Assessments_Term2' and 'Course_Credit_Master'. The 'Flow' canvas shows a data flow from 'Student_Assessments_Term2' to 'Clean 1', then to 'Clean 2'. The 'Clean 2' step is highlighted. Below the flow, a preview table is shown with the following data:

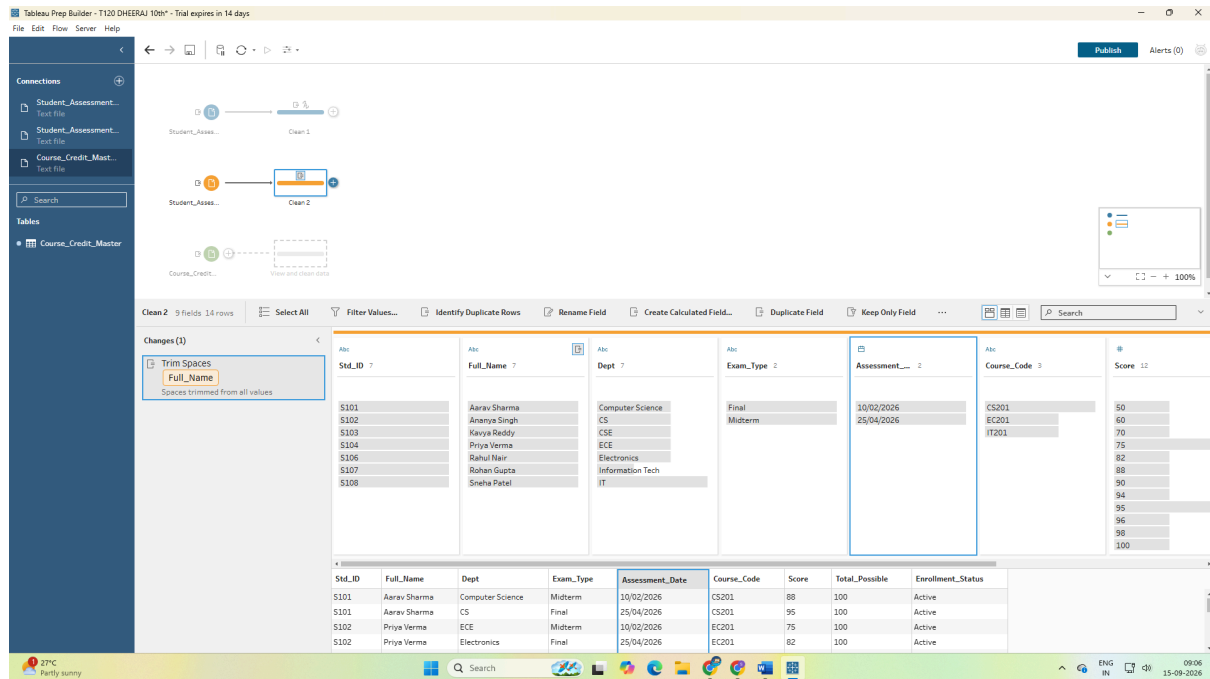
Std_ID	Full_Name	Dept	Exam_Type	Assessment_Date	Course_Code	Score	Total_Possible	Enrollment_Status
S101	Aarav Sharma	Computer Science	Midterm	10/02/2026	CS201	88	100	Active
S101	Aarav Sharma	CS	Final	25/04/2026	CS201	95	100	Active
S102	Priya Verma	ECE	Midterm	10/02/2026	EC201	75	100	Active
S102	Priya Verma	Electronics	Final	25/04/2026	EC201	82	100	Active

Note: Tableau Prep may have already Selected the date datatype.

Step 5: Review the Changes Pane

1. Look at the left side of the screen under the Changes tab.
2. Notice that Tableau Prep records every single transformation as an itemized, reversible step (e.g., Trim spaces on Student_Name, Change data type of Exam_Date).

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C: Group and Standardize Values

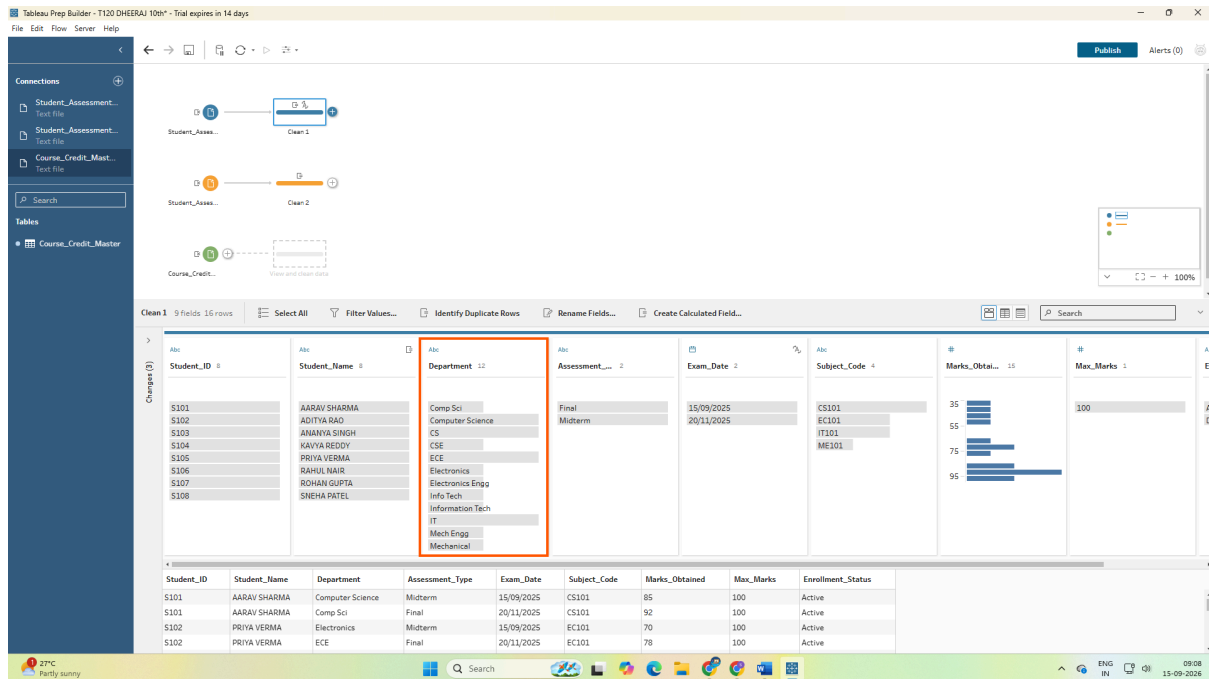
Objective: Standardize spelling variations, abbreviations, and synonyms in categorical columns (such as Department and Dept) into clean, consistent values using Tableau Prep's built-in grouping algorithms.

In academic datasets, departments are frequently entered under different aliases (e.g., "Comp Sci", "CS", "CSE", and "Computer Science"). If left unstandardized, downstream visual analysis will treat each alias as a completely different academic department, splintering the summary metrics.

Step 1: Inspect the Department Field in Term 1 (Clean 1)

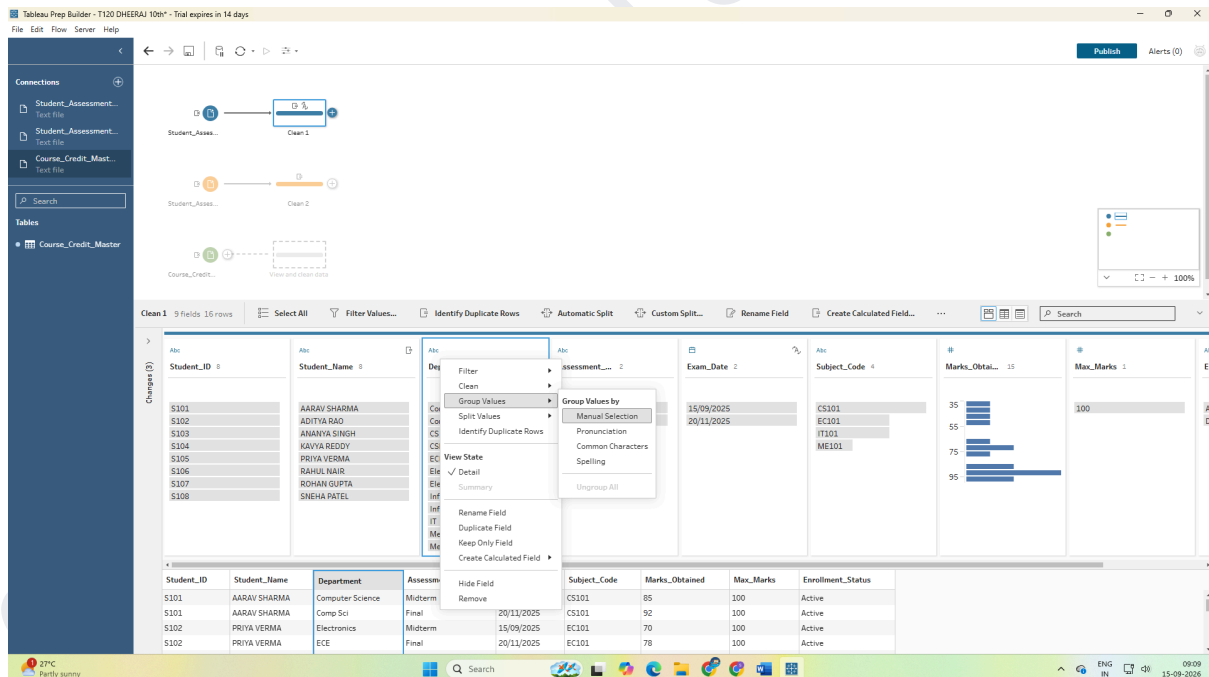
1. Click on the Clean 1 step connected to Student_Assessments_Term1.
2. Locate the Department card in the Profile Pane.
3. Observe that there are 12 distinct entries representing only 4 actual academic departments:
 - o Computer Science (Computer Science, Comp Sci, CS, CSE)
 - o Electronics (Electronics, ECE, Electronics Engg)
 - o Information Technology (Information Tech, IT, Info Tech)
 - o Mechanical Engineering (Mechanical, Mech Engg)

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Step 2: Group Values Using Pronunciation / Spelling or Manual Selection1. On the Department card, click the More options (...) menu at the top right.

2. Go to Group Values -->select Manual Selection (or Pronunciation / Common Characters).



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3. In the Grouping editor: o Click on Computer Science in the left column. In the right-hand panel, check the boxes for Comp Sci, CS, and CSE.

The screenshot shows the Tableau Prep Builder interface. In the 'Group Values by Manual Selection' panel, 'Computer Science' is selected in the left column. In the right column, the checkboxes for 'Comp Sci', 'CS', and 'CSE' are checked. The data table below shows the following rows:

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S101	AARAV SHARMA	Computer Science	Midterm	15/09/2025	CS101	85	100	Active
S103	ROHAN GUPTA	Computer Science	Midterm	15/09/2025	CS101	45	100	Active
S103	ROHAN GUPTA	Computer Science	Final	20/11/2025	CS101	52	100	Active
S106	ANANYA SINGH	Computer Science	Midterm	15/09/2025	CS101	88	100	Active

o Click on Electronics (or rename the master value to Electronics & Communication).

Check the boxes for ECE and Electronics Engg.

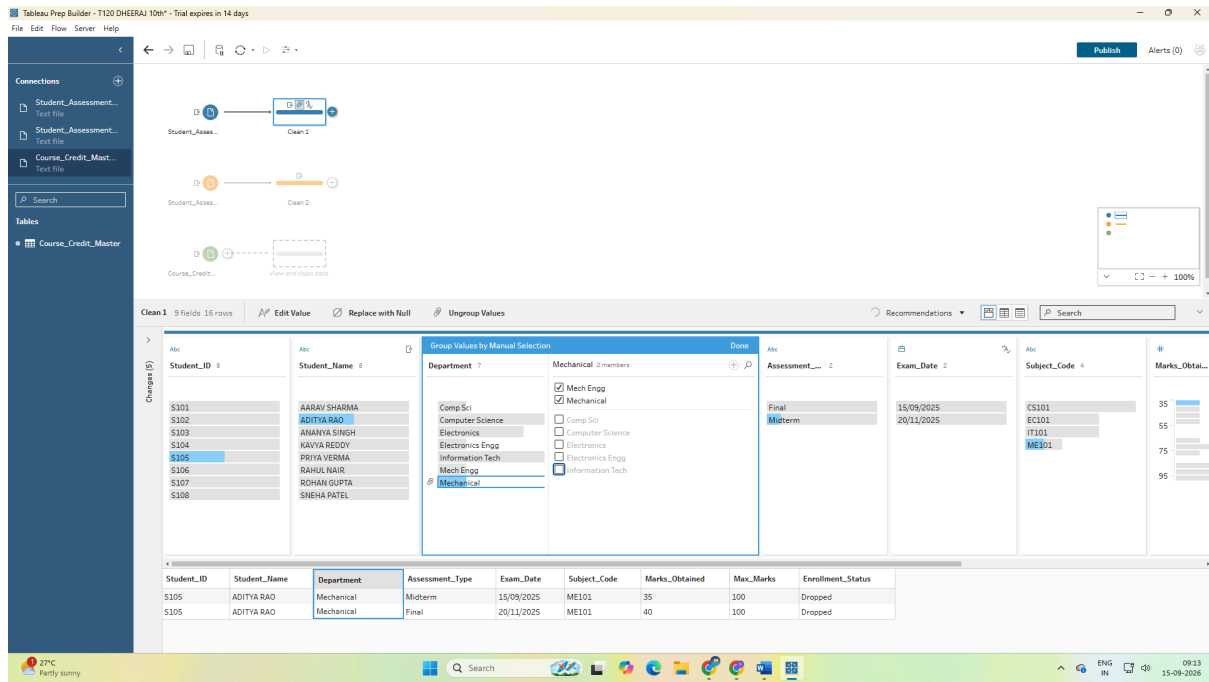
The screenshot shows the Tableau Prep Builder interface. In the 'Group Values by Manual Selection' panel, 'Electronics' is selected in the left column. In the right column, the checkboxes for 'ECE' and 'Electronics Engg' are checked. The data table below shows the following rows:

Student_ID	Student_Name	Department	Assessment_Type	Exam_Date	Subject_Code	Marks_Obtained	Max_Marks	Enrollment_Status
S102	PRIYA VERMA	Electronics	Midterm	15/09/2025	EC101	70	100	Active
S102	PRIYA VERMA	Electronics	Final	20/11/2025	EC101	78	100	Active
S107	RAHUL NAIR	Electronics	Midterm	15/09/2025	EC101	65	100	Active

o Click on Information Tech (or rename to Information Technology).

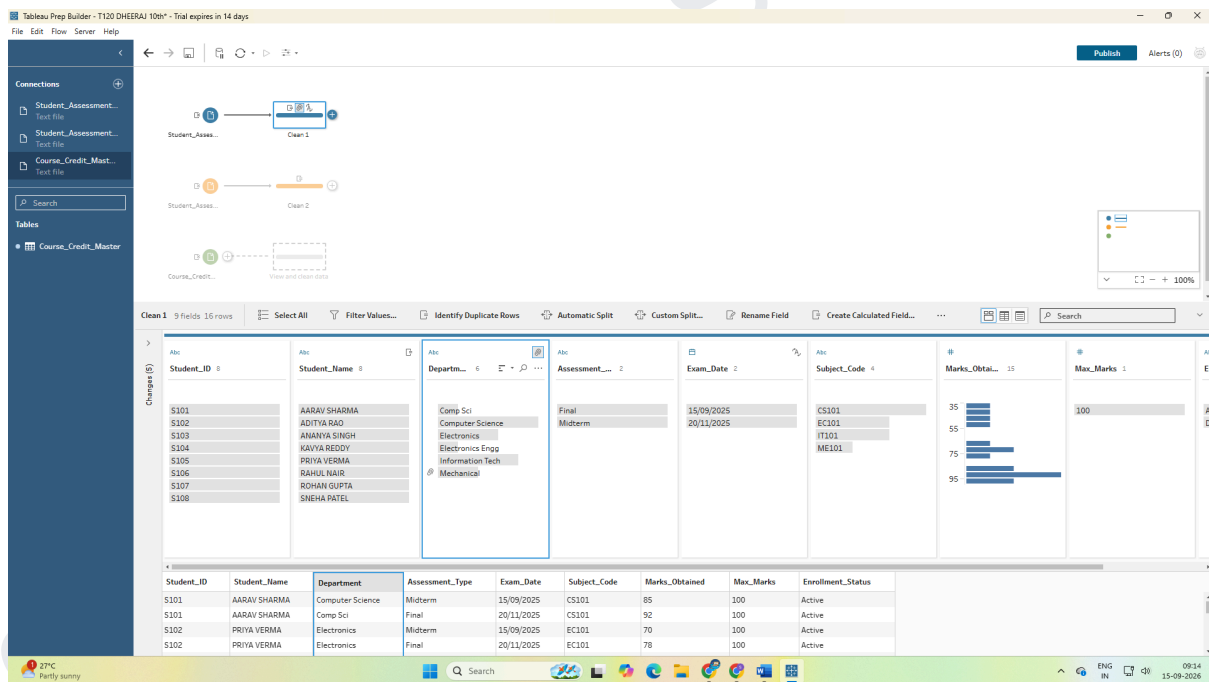
Check the boxes for IT and Info Tech.

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o Click on Mechanical (or rename to Mechanical Engineering). Check the box for Mech Engg.

4. Click Done at the top right of the editor.

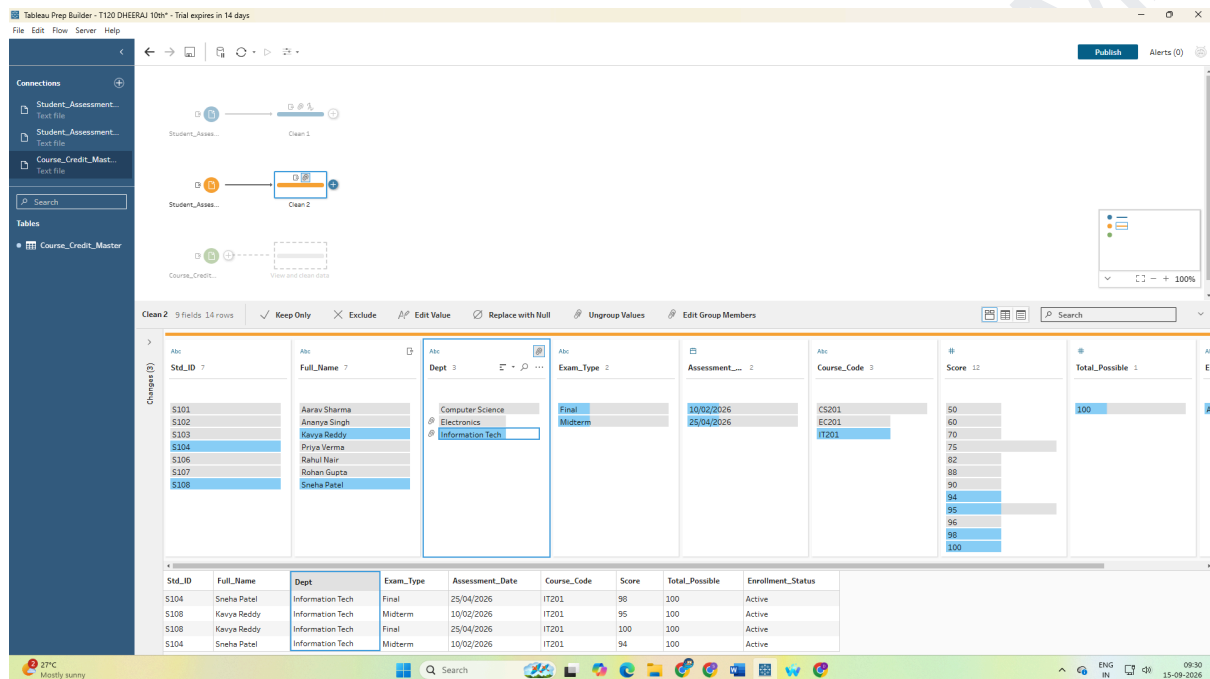


5. Notice that the paperclip icon appears next to grouped values, consolidating the 12 messy entries into exactly 4 clean departments

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Step 3: Repeat Grouping on Term 2 (Clean 2)

1. Click on the Clean 2 step connected to Student_Assessments_Term2.
2. Locate the Dept card.
3. Click More options (...) -->Group Values -->Manual Selection:
 - o Group CS, CSE, and Computer Science into Computer Science.
 - o Group ECE and Electronics into Electronics.
 - o Group IT and Information Tech into Information Tech.
4. Click Done.



D: Merge Mismatched Fields

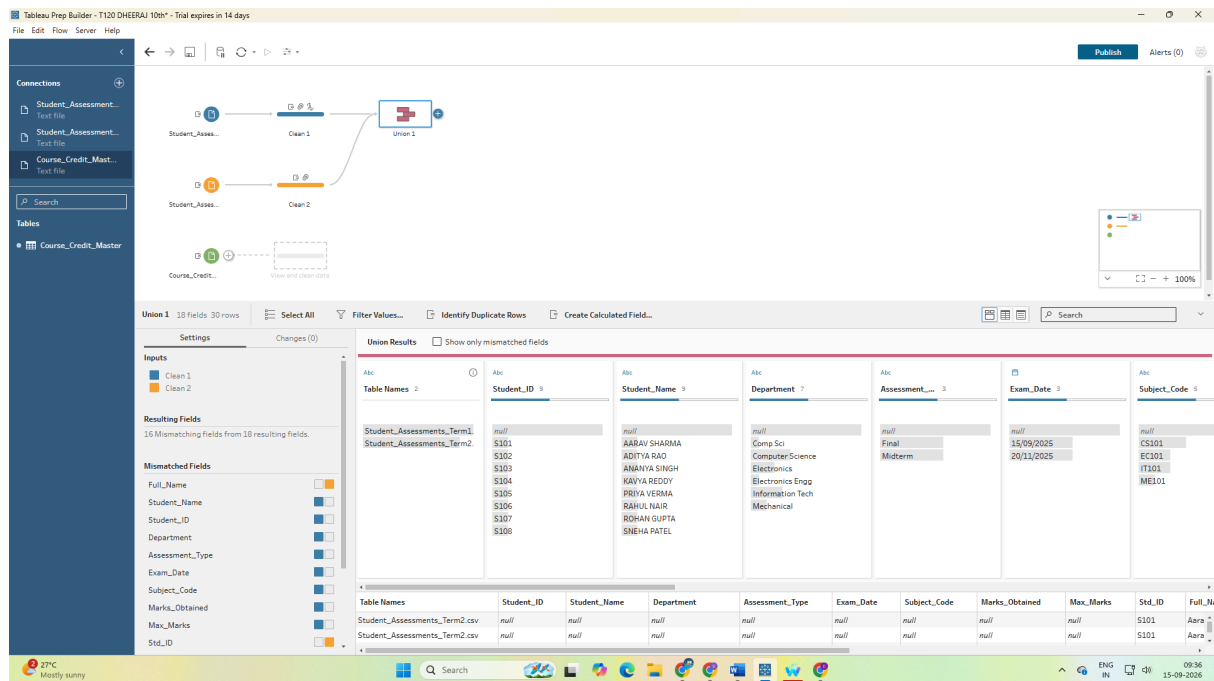
Objective: Perform a vertical Union of Clean 1 (Term 1) and Clean 2 (Term 2), resolve mismatched column headers (such as Std_ID vs Student_ID, Score vs Marks_Obtained), and unify them into a single schema.

When combining datasets originating from different semesters, departments, or source systems, identical information often resides under dissimilar column names. In Tableau Prep, unmerged fields create sparse datasets with duplicate null columns. Merging mismatched fields consolidates those columns without requiring upstream source file modifications.

Step 1: Create the Union Step on the Canvas

1. On the Flow canvas, click and drag the Clean 2 step icon over to the Clean 1 step icon.
2. When the drop zone appears, hover over Union and release the mouse.
3. A new Union 1 step is created, connecting both transformation branches.

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Step 2: Inspect Mismatched Fields in the Union Pane

1. Click on the Union 1 step.
2. In the bottom Union summary pane, look at the Mismatched Fields section.
3. Notice that because the two files used different naming conventions, Tableau Prep lists non-matching pairs (indicated by yellow and blue bars), such as:
 - o Student_ID (Term 1) vs Std_ID (Term 2)
 - o Student_Name (Term 1) vs Full_Name (Term 2)
 - o Department (Term 1) vs Dept (Term 2)
 - o Assessment_Type (Term 1) vs Exam_Type (Term 2)
 - o Exam_Date (Term 1) vs Assessment_Date (Term 2)
 - o Subject_Code (Term 1) vs Course_Code (Term 2)
 - o Marks_Obtained (Term 1) vs Score (Term 2)
 - o Max_Marks (Term 1) vs Total_Possible (Term 2)

Step 3: Merge the Mismatched Fields

1. In the Mismatched Fields list (or the union profile pane):
 - o Click and drag Std_ID directly on top of Student_ID to merge them into one column.
 - o Drag Full_Name onto Student_Name.
 - o Drag Dept onto Department.

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Note: If You cant drag it Observe in the left pane click the dept column the hover near department column's plus icon and add the dept column into it o Drag Exam_Type onto Assessment_Type.

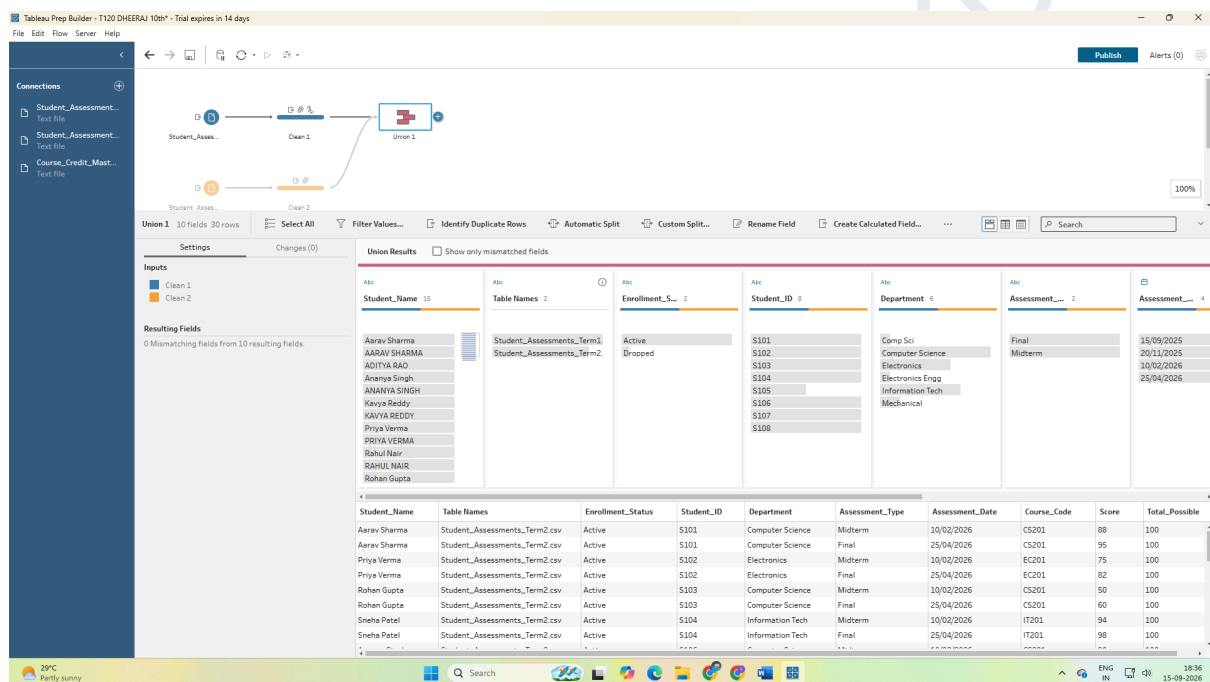
o Drag Assessment_Date onto Exam_Date.

o Drag Course_Code onto Subject_Code.

o Drag Score onto Marks_Obtained.

o Drag Total_Possible onto Max_Marks.

2. Notice that as you merge them, the mismatched count drops to 0, and each unified column displays both source colors.



Step 4: Add a Clean Step Post-Union 1. Click the + icon next to the Union 1 node.

2. Select Clean Step (creates Clean 3). In Screenshot it is showing Clean 4 because I have created clean step for Course_Credit_Master.csv too.

3. Verify that the combined dataset now contains 30 total rows across 10 unified columns (including Table Names).

E: Perform Aggregations in Prep

Objective: Create an Aggregate Step in Tableau Prep to summarize granular, exam-level test scores into student-level cumulative metrics (Total Marks, Average Score, and Total Assessment Count) per term.

Raw assessment tables log every individual examination (Midterm, Final) on a separate row. By performing an aggregation step directly in Tableau Prep, we roll up individual row events into

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pre-computed summary statistics, reducing data redundancy and generating student scorecard baselines prior to dashboarding.

Step 1: Add a Term Identifier Field in Clean 3

1. Click on the Clean 3 step (the clean step created after Union 1).
2. In the toolbar above the Profile cards, click Create Calculated Field You have to click three dots on the toolbar.

Student Name	Table Names	Enrollment Status	Student ID	Department	Assessment Type	Assessment Date	Course Code	Score	Total Possible
Aarav Sharma	Student_Assessments_Term1.csv	Active	S101	Computer Science	Midterm	10/02/2026	CS201	88	100
Aarav Sharma	Student_Assessments_Term2.csv	Active	S101	Computer Science	Final	25/04/2026	CS201	95	100
Priya Verma	Student_Assessments_Term2.csv	Active	S102	Electronics	Midterm	10/02/2026	EC201	75	100
Priya Verma	Student_Assessments_Term2.csv	Active	S102	Electronics	Final	25/04/2026	EC201	82	100
Rohan Gupta	Student_Assessments_Term2.csv	Active	S103	Computer Science	Midterm	10/02/2026	CS201	50	100
Rohan Gupta	Student_Assessments_Term2.csv	Active	S103	Computer Science	Final	25/04/2026	CS201	60	100
Sneha Patel	Student_Assessments_Term2.csv	Active	S104	Information Tech	Midterm	10/02/2026	IT201	94	100
Sneha Patel	Student_Assessments_Term2.csv	Active	S104	Information Tech	Final	25/04/2026	IT201	98	100
Ananya Singh	Student_Assessments_Term2.csv	Active	S106	Computer Science	Midterm	10/02/2026	CS201	90	100
Ananya Singh	Student_Assessments_Term2.csv	Active	S106	Computer Science	Final	25/04/2026	CS201	96	100

3. Set the Field Name to: Term

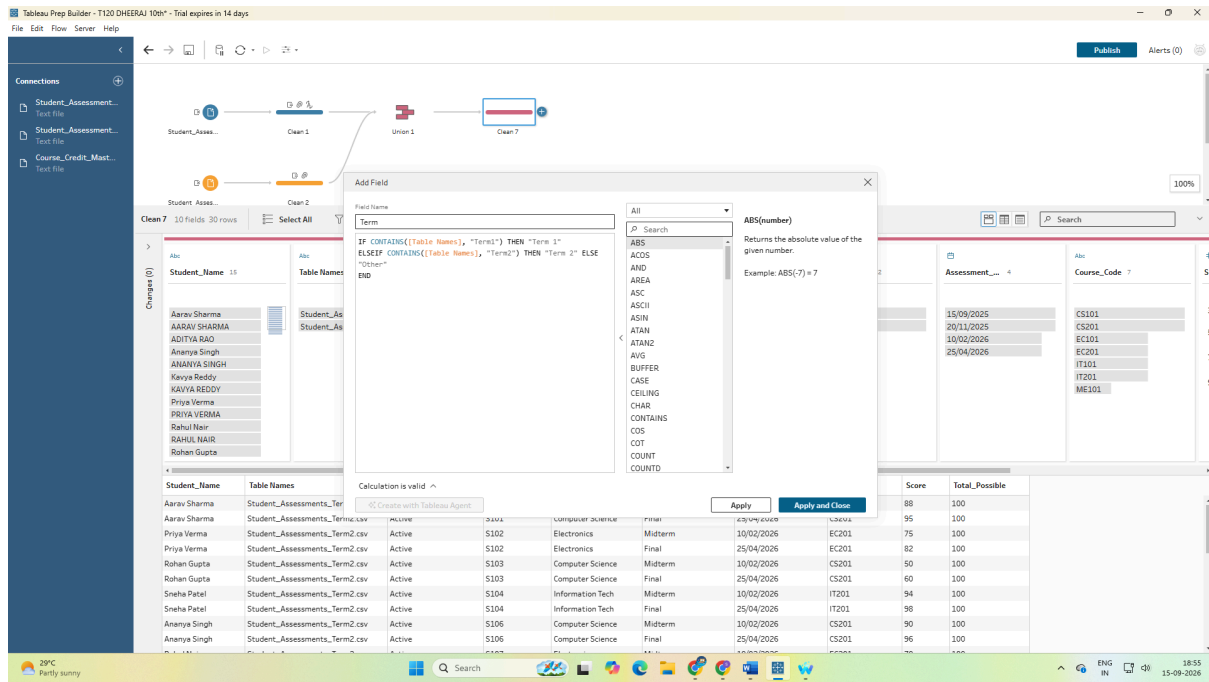
4. Enter the formula:

Code snippet

```
IF CONTAINS([Table Names], "Term1") THEN "Term 1"
ELSEIF CONTAINS([Table Names], "Term2") THEN "Term 2"
ELSE "Other"
END
```

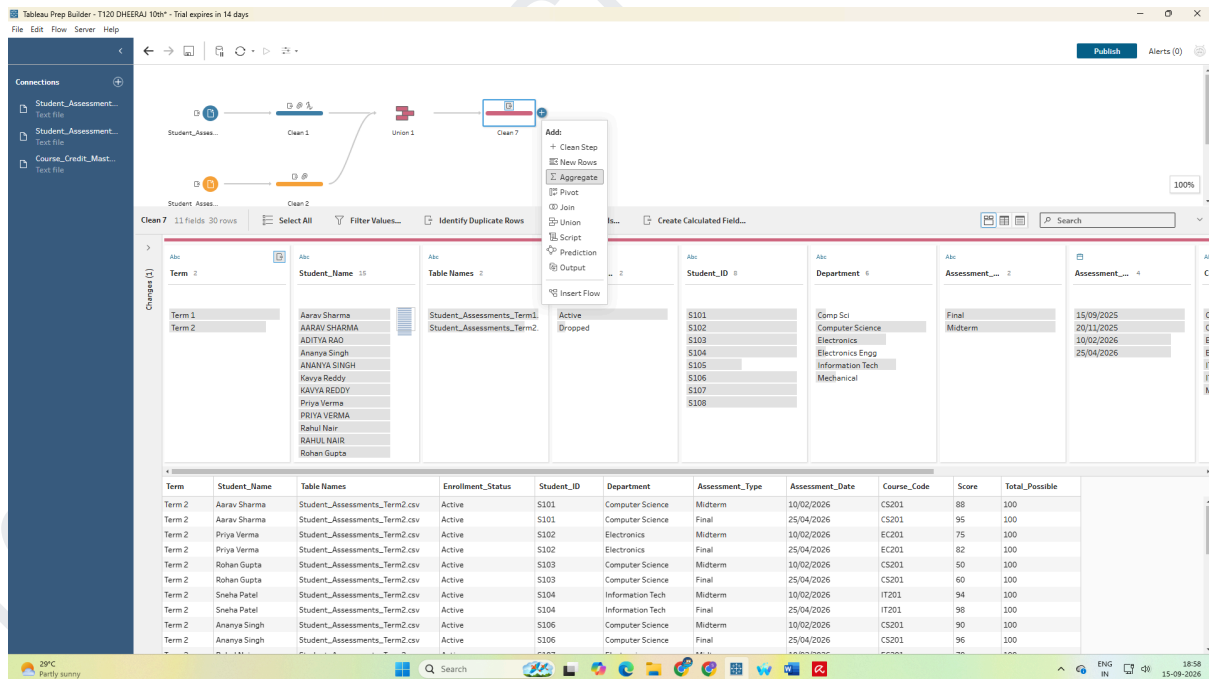
5. Click Save.

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Step 2: Add an Aggregate Step to the Flow Canvas

1. Hover over the Clean 3 node on the Flow canvas.
2. Click the + icon next to it.
3. Select Aggregate (represented by the summation icon Σ).



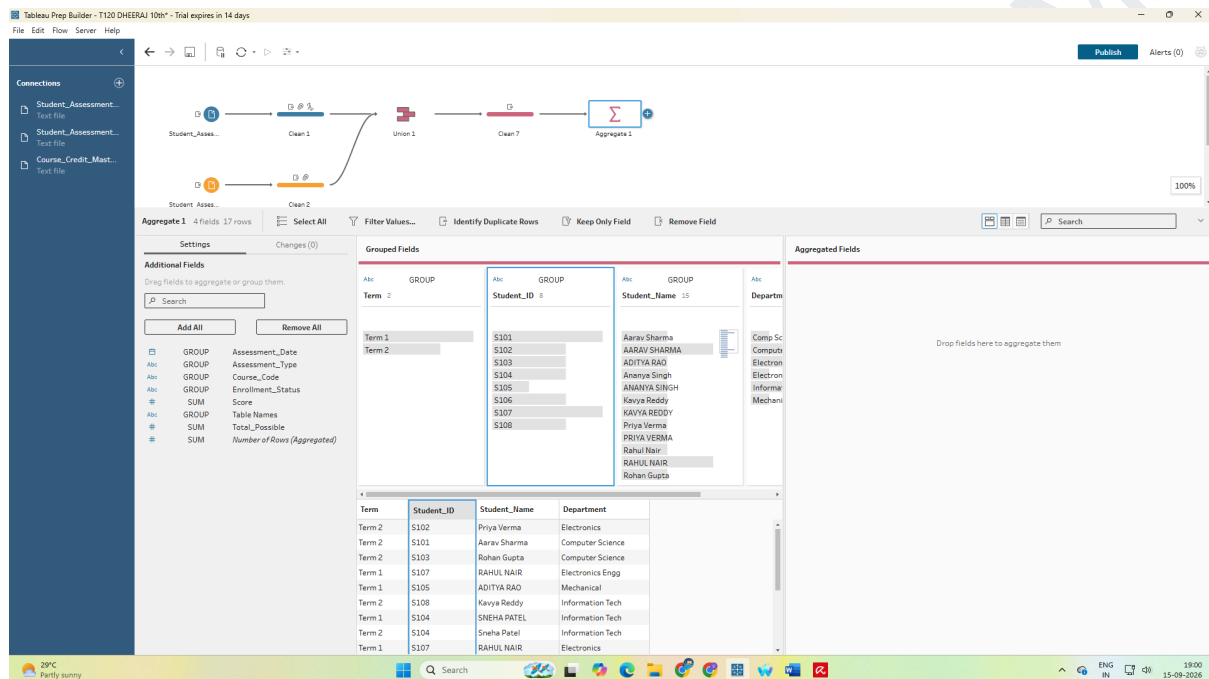
4. An Aggregate 1 step will appear, showing two drop zones in the configuration pane:

- o Grouped Fields (Left Zone)
- o Aggregated Fields (Right Zone)

Step 3: Configure Grouped Fields (Dimensions)

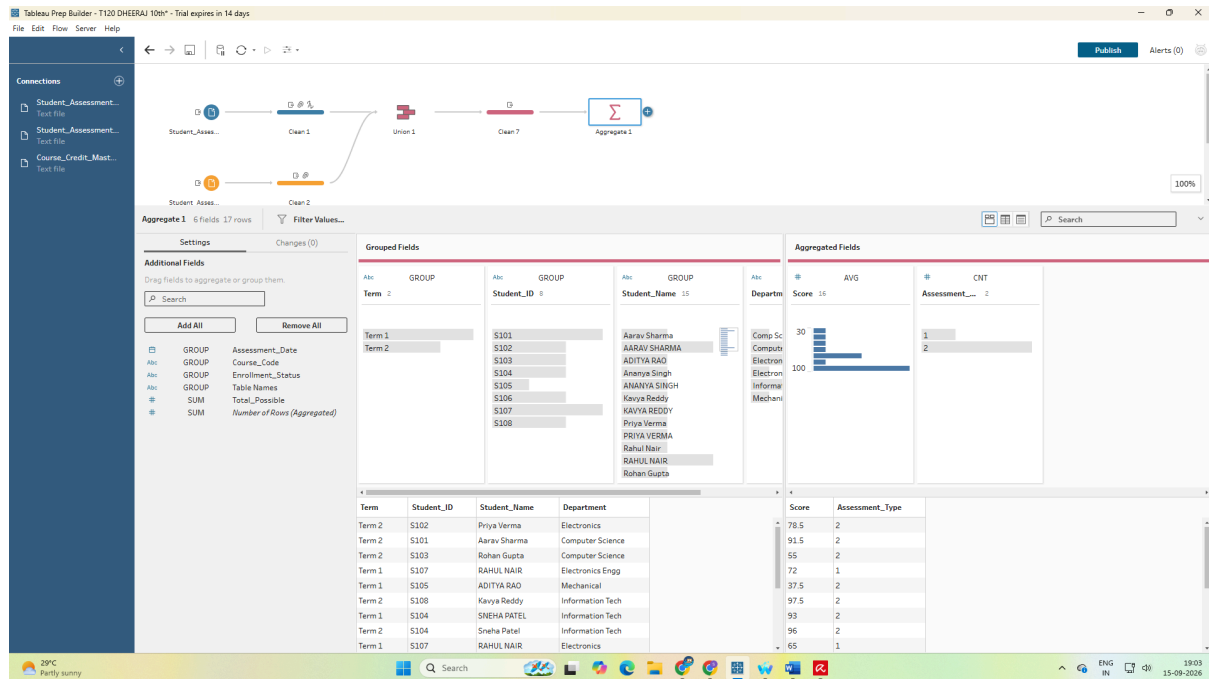
From the left-hand column list in the Aggregate step pane Your Column names could be different based on how you did previous steps of Union:

1. Drag Student_ID into the Grouped Fields box.
2. Drag Student_Name into the Grouped Fields box.
3. Drag Department into the Grouped Fields box.
4. Drag Term into the Grouped Fields box.



Step 4: Configure Aggregated Fields (Measures & Metrics) 1. Total Marks: Drag Marks_Obtained into the Aggregated Fields box. (Ensure aggregation operation is set to Sum). Data Visualization with

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2. Average Score: Drag Marks_Obtained into the Aggregated Fields box a second time --> click on Sum --> change it to Average. There will be no field for dragging by any chance just change SUM to AVERAGE of the existing one.
 3. Total Possible Marks: Drag Max_Marks into the Aggregated Fields box --> keep as Sum.
 4. Assessment Count: Drag Assessment_Type (or Subject_Code) into the Aggregated Fields box --> set aggregation to Count (or Count Distinct)
- Step

5: Rename Aggregated Output Fields

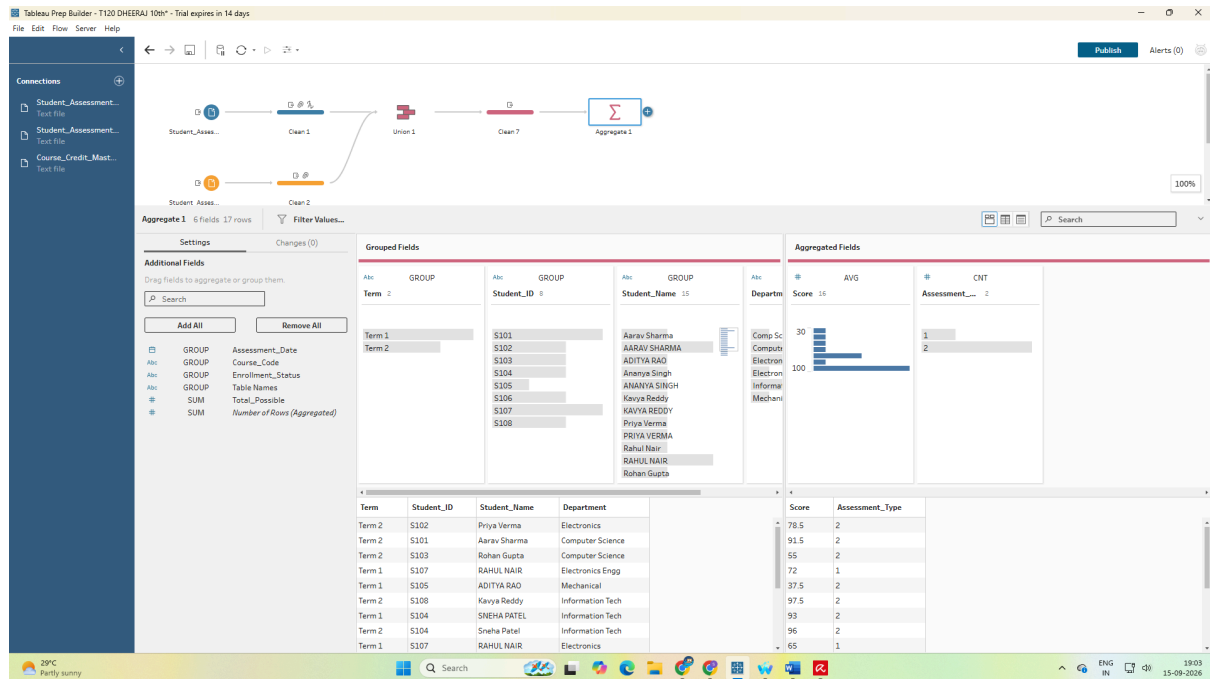
In the Aggregated Fields panel, click on each field name to rename them cleanly:

- Marks_Obtained (Sum) --> Total_Marks_Obtained
- Marks_Obtained (Average) --> Average_Score
- Max_Marks (Sum) --> Total_Max_Marks
- Assessment_Type (Count) --> Assessments_Taken

Step 6: Add a Clean Step to Inspect Aggregated Output

1. Click the + icon next to the Aggregate 1 node --> select Clean Step (creates Clean 4).
2. Look at the Profile Pane: o The row count collapses from 30 individual assessment rows into 15 summary records or more (one row per student per term).

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F: Apply Filters and Transformations

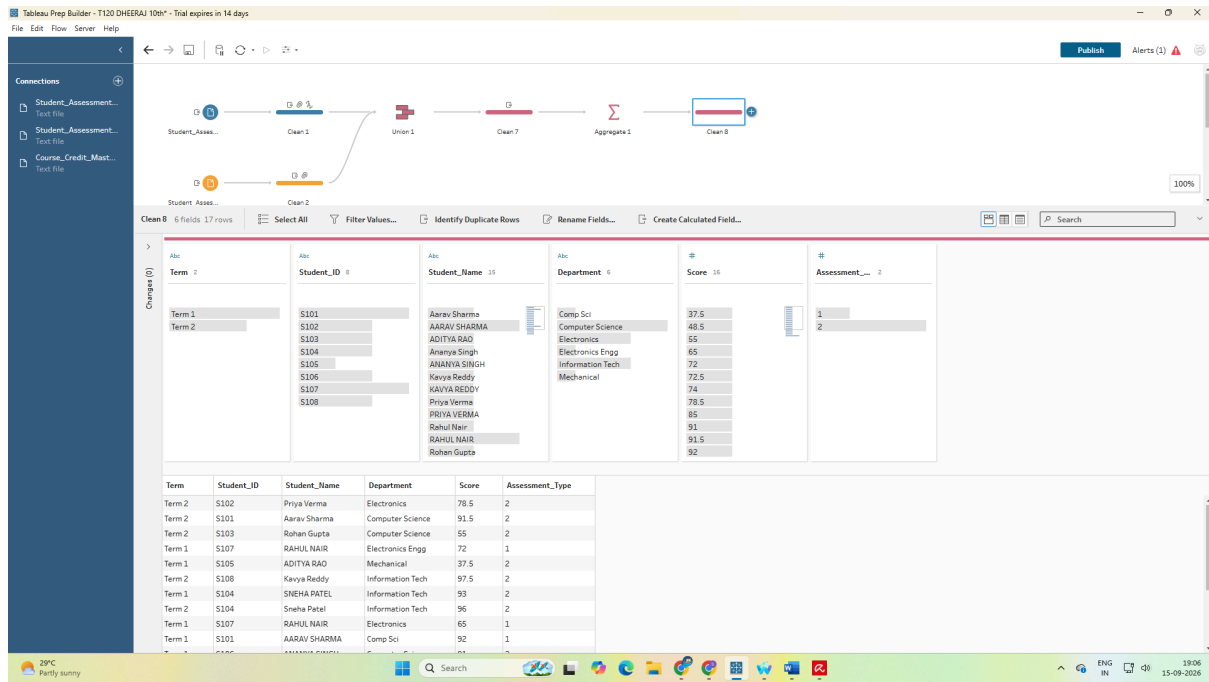
Objective: Remove inactive/dropped records, compute weighted percentages, assign automated grading tiers, and derive academic standing indicators using calculated fields in Tableau Prep.

Not all raw records should enter downstream analytical dashboards (e.g., students who dropped out mid-term skew department averages). Transforming raw totals into percentage metrics and standardized letter grades directly inside the data prep pipeline ensures that business logic is applied once and remains consistent across all downstream reporting tools.

Step 1: Open the Transformation Step (Clean 4)

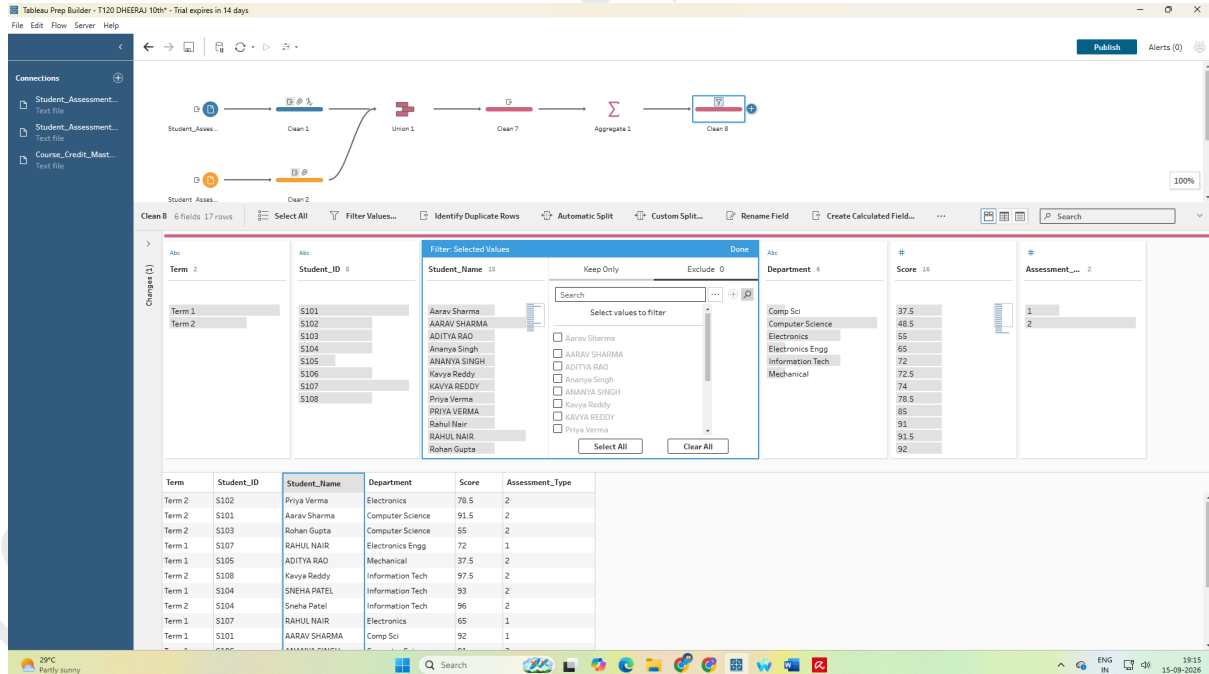
1. Click on the Clean 4 step connected after your Aggregate 1 step on the Flow canvas.
2. The Profile Pane below will show your aggregated columns: Student_ID, Student_Name, Department, Term, Total_Marks_Obtained, Average_Score, Total_Max_Marks, and Assessments_Taken. (Naming Will be different and completely depends on how you performed the Union Step)

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Step 2: Filter Out Inactive / Dropped Student Records

1. If Enrollment_Status is present on a profile card: o Click on the card --> click the value Dropped --> right-click and select Exclude (or click the filter icon at the top of the card and check only Active).
2. (Alternative via Filter calculation): o In the top toolbar, click Filter Values --> Selected Values....



Step 3: Create the Normalized Term Percentage Metric

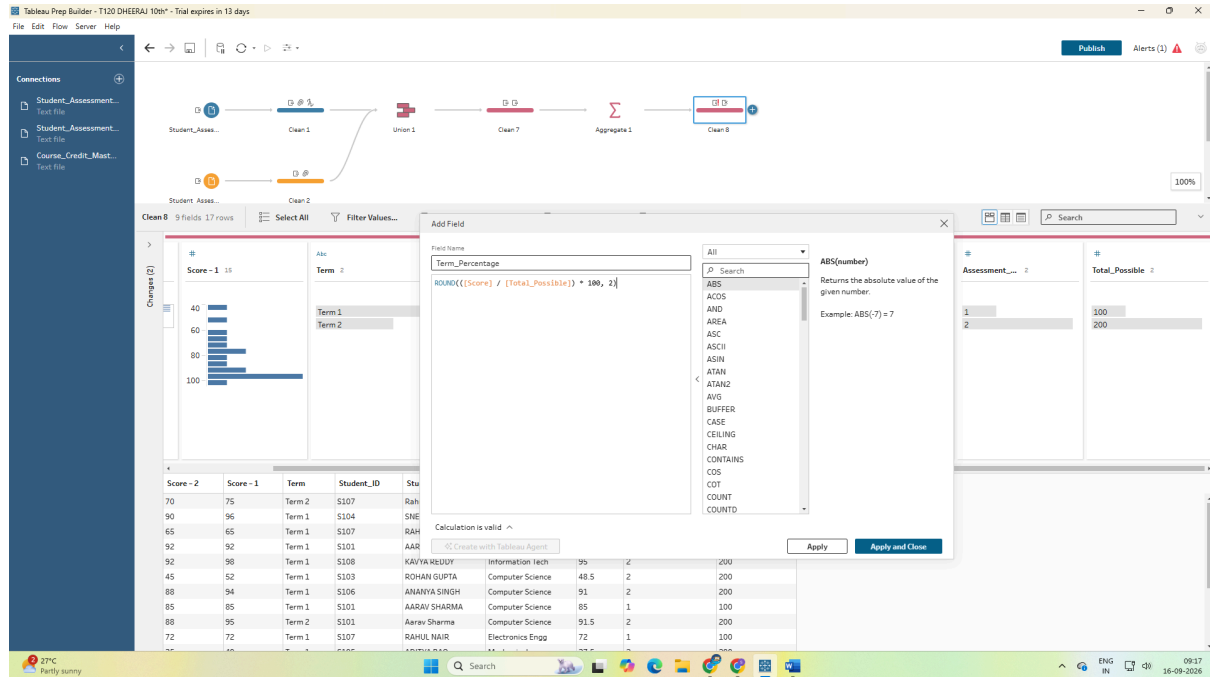
1. In the toolbar above the Profile cards, click Create Calculated Field.
2. Field Name: Term_Percentage
3. Tableau Prep Formula:

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Code snippet :

$\text{ROUND}([\text{Score}] / [\text{Total_Possible}] * 100, 2)$ {This formula can be used if you couldn't able to rename your aggregate fields.}

4. Click Save.



Step 4: Create the Categorical Letter Grade Transformation

1. In the toolbar, click Create Calculated Field.

2. Field Name: Letter_Grade

3. Tableau Prep Formula:

Code snippet

IF [Term_Percentage] >= 85 THEN "A (Distinction)"

ELSEIF [Term_Percentage] >= 75 THEN "B (First Class)"

ELSEIF [Term_Percentage] >= 60 THEN "C (Second Class)"

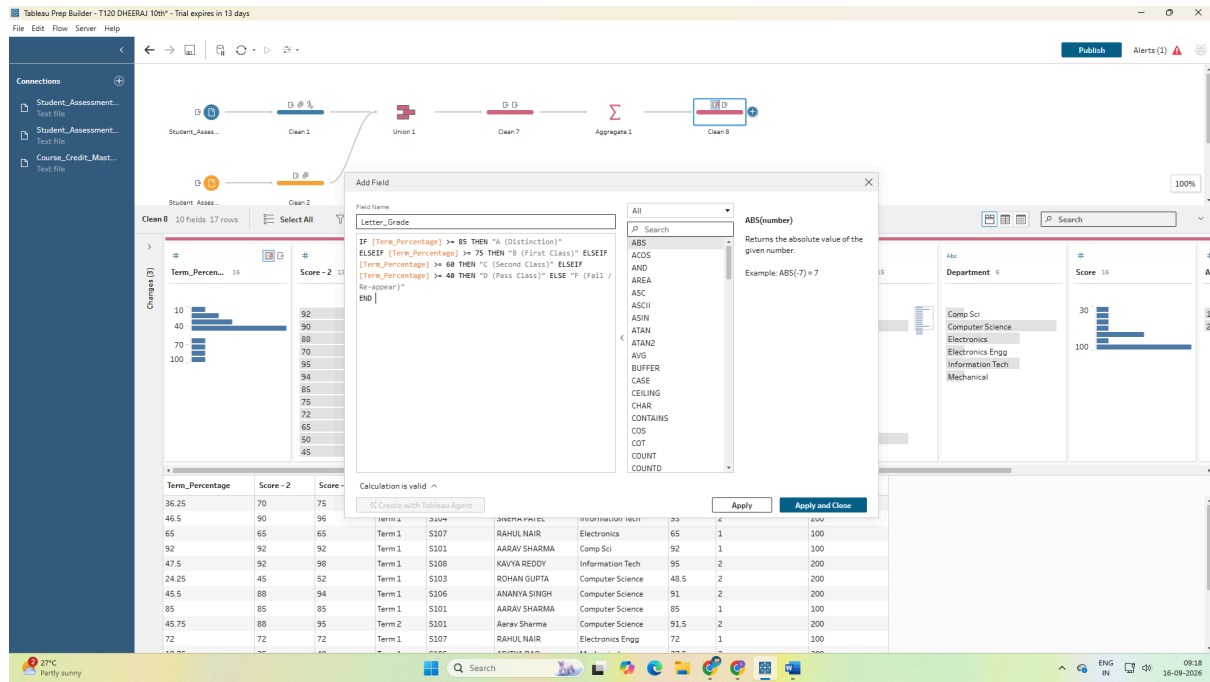
ELSEIF [Term_Percentage] >= 40 THEN "D (Pass Class)"

ELSE "F (Fail / Re-appear)"

END

4. Click Save.

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Step 5: Create Academic Standing Compliance Flag

1. Click Create Calculated Field.
2. Field Name: Academic_Standing
3. Tableau Prep Formula:

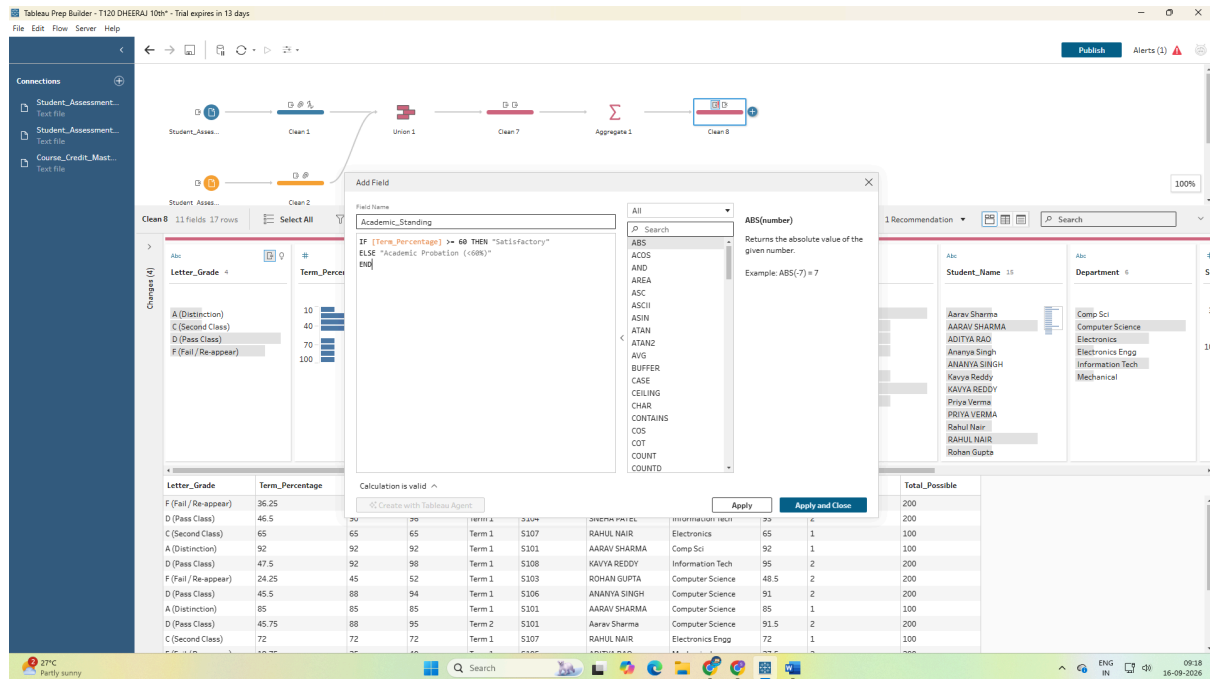
Code snippet

IF [Term_Percentage] >= 60 THEN "Satisfactory"

ELSE "Academic Probation (<60%)"

END

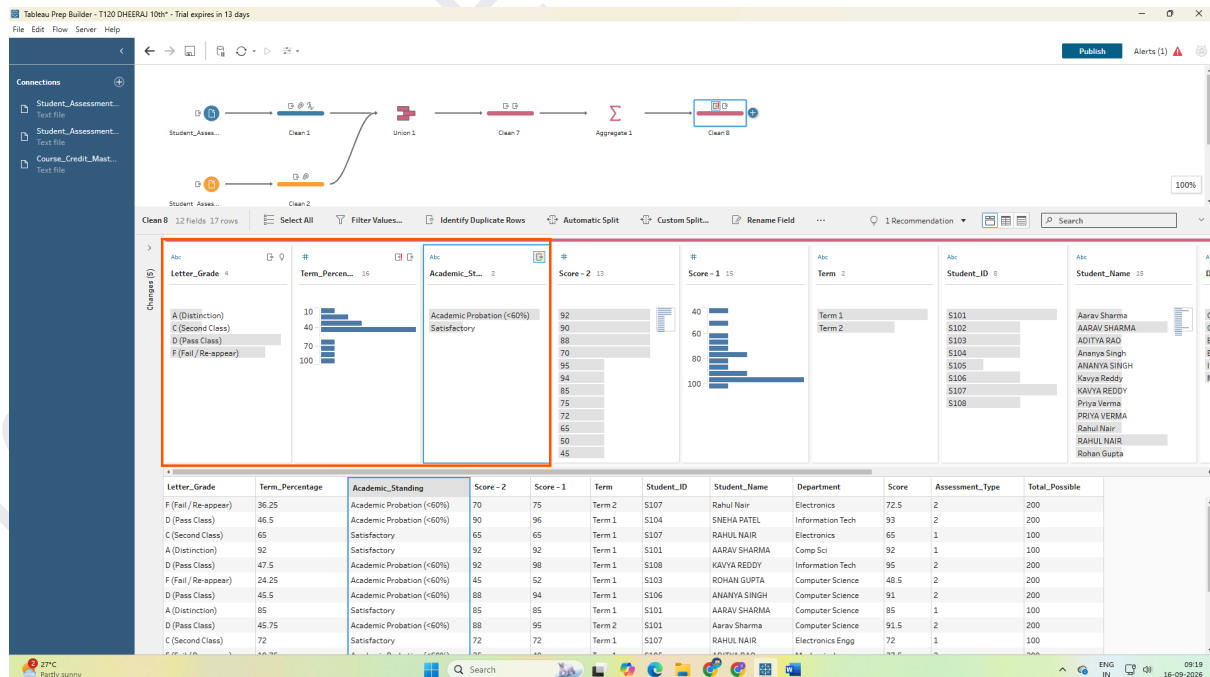
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4. Click Save.

Step 6: Verify Transformed Profile Distributions

- Look at the newly generated profile cards in Clean 4:
 - Term_Percentage: Displays a smooth numeric distribution between 0% and 100%.
 - Letter_Grade: Displays discrete categorical buckets (A (Distinction), B (First Class), etc.).
 - Academic_Standing: Highlights students requiring academic counseling.



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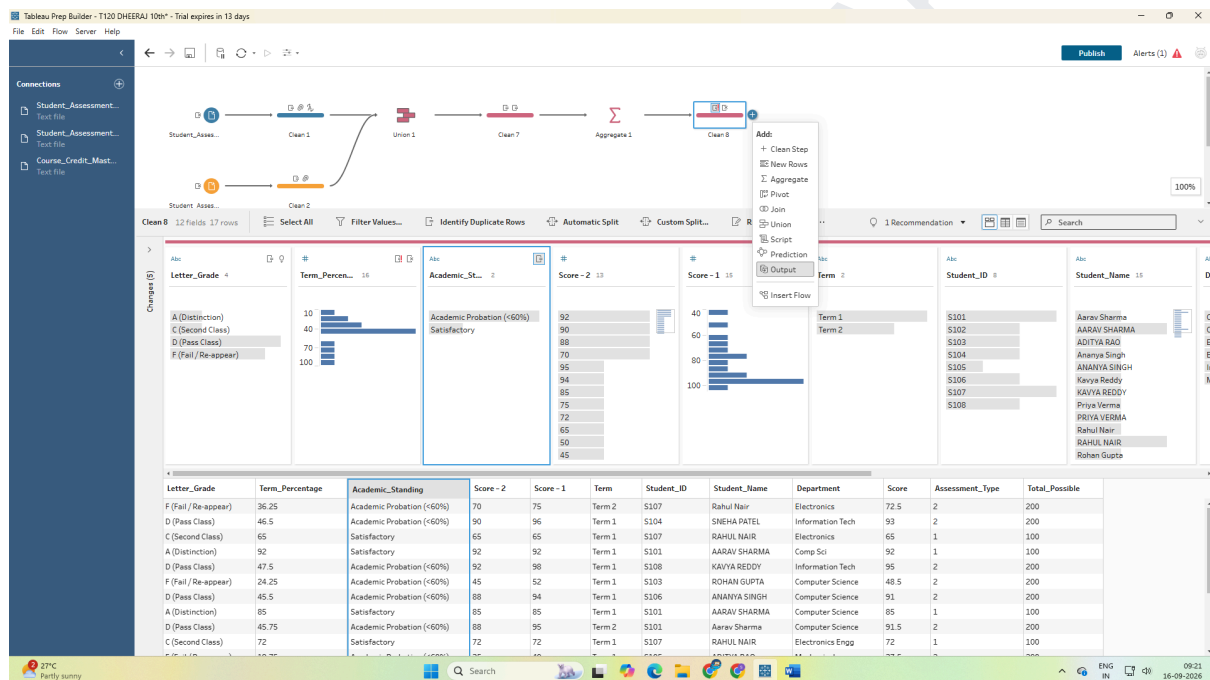
G: Create and Automate Data Preparation Flows

Objective: Configure an Output Step in Tableau Prep Builder to publish the fully transformed pipeline to an extract file (.hyper or .csv), and learn how to automate or execute the flow on demand.

Data preparation workflows must be repeatable and production-ready. Configuring a dedicated output node ensures that whenever new assessment scores arrive, running the flow automatically executes all trimming, grouping, merging, filtering, and scoring calculations in seconds without any manual rework.

Step 1: Add an Output Step to the End of the Pipeline

1. On the Flow canvas, locate the final cleaning step (Clean 4).
2. Hover over the Clean 4 icon and click the + icon.
3. Select Output (represented by the download/save arrow icon).
4. An Output step node will appear connected directly to Clean 4.



Step 2: Configure Output Settings in the Bottom Pane

1. Click on the Output node to open its configuration panel below:
 - o Save output to: Select File (or Tableau Server / Tableau Cloud).
 - o Output type: Select Tableau Data Extract (.hyper) for high-speed dashboarding, or Comma Delimited Text (.csv) for universal export.
 - o Location: Click Browse and choose the directory where you want the final dataset saved.
 - o File Name: Enter Student_Performance_Master_Cleaned.
 - o Write Option: Select Create table (or Append to table if updating existing records).

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Tableau Prep Builder - T120 DHEERAJ 10th* - Trial expires in 13 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Student_Assessment... Clean 1

Union 1

Clean 7

Aggregate 1

Clean 8

Output

Output: 12 Fields

Save output to: File

Write Options: Create table

Run Flow

Letter_Grade	Term_Percentage	Academic_Standing	Score - 2	Score - 1	Term	Student_ID	Student_Name	Department	Score	Assessment_Type	Total
F (Fail / Re-appear)	36.25	Academic Probation (<60%)	70	75	Term 2	S107	Rahul Nair	Electronics	72.5	2	200
D (Pass Class)	46.5	Academic Probation (<60%)	90	96	Term 1	S104	SNEHA PATEL	Information Tech	93	2	200
C (Second Class)	65	Satisfactory	65	65	Term 1	S107	RAHUL NAIR	Electronics	65	1	100
A (Distinction)	92	Satisfactory	92	92	Term 1	S101	AARAV SHARMA	Comp Sci	92	1	100
D (Pass Class)	47.5	Academic Probation (<60%)	92	98	Term 1	S108	KAVYA REDDY	Information Tech	95	2	200
F (Fail / Re-appear)	24.25	Academic Probation (<60%)	45	62	Term 1	S103	ROHAN GUPTA	Computer Science	48.5	2	200
D (Pass Class)	45.5	Academic Probation (<60%)	88	94	Term 1	S106	ANANYA SINGH	Computer Science	91	2	200
A (Distinction)	85	Satisfactory	85	85	Term 1	S101	AARAV SHARMA	Computer Science	85	1	100
D (Pass Class)	45.75	Academic Probation (<60%)	88	95	Term 2	S101	Aarav Sharma	Computer Science	91.5	2	200
C (Second Class)	72	Satisfactory	72	72	Term 1	S107	RAHUL NAIR	Electronics Engg	72	1	100
F (Fail / Re-appear)	18.75	Academic Probation (<60%)	35	40	Term 1	S105	ADITYA RAO	Mechanical	37.5	2	200
F (Fail / Re-appear)	39.25	Academic Probation (<60%)	75	82	Term 2	S102	Priya Verma	Electronics	78.5	2	200
F (Fail / Re-appear)	37	Academic Probation (<60%)	70	78	Term 1	S102	PRIYA VERMA	Electronics	74	2	200
D (Pass Class)	48.75	Academic Probation (<60%)	95	100	Term 2	S108	Kavya Reddy	Information Tech	97.5	2	200
F (Fail / Re-appear)	27.5	Academic Probation (<60%)	50	60	Term 2	S103	Rohan Gupta	Computer Science	55	2	200
D (Pass Class)	46.5	Academic Probation (<60%)	90	96	Term 2	S106	Ananya Singh	Computer Science	93	2	200
D (Pass Class)	48	Academic Probation (<60%)	94	98	Term 2	S104	Sneha Patel	Information Tech	96	2	200

Step 3: Run the Data Prep Flow

1. In the Output pane at the bottom-left, click the blue Run Flow button (or click the play icon  directly on the Output node on the canvas).
2. Tableau Prep will process all upstream nodes (Cleaning $\rightarrow$ Grouping $\rightarrow$ Unioning $\rightarrow$ Merging $\rightarrow$ Aggregating $\rightarrow$ Scoring).
3. A success notification will appear stating:
"The flow ran successfully in X seconds."

Tableau Prep Builder - T120 DHEERAJ 10th* - Trial expires in 13 days

File Edit Flow Server Help

Connections

- Student_Assessment... Text file
- Student_Assessment... Text file
- Course_Credit_Mast... Text file

Student_Assessment... Clean 1

Union 1

Clean 7

Aggregate 1

Clean 8

Output

Output: 12 Fields

Save output to: File

Write Options: Create table

Run Flow

Finished Running Flow

Student_Performance_Master_Cleaned.hyper

Total time 00:00

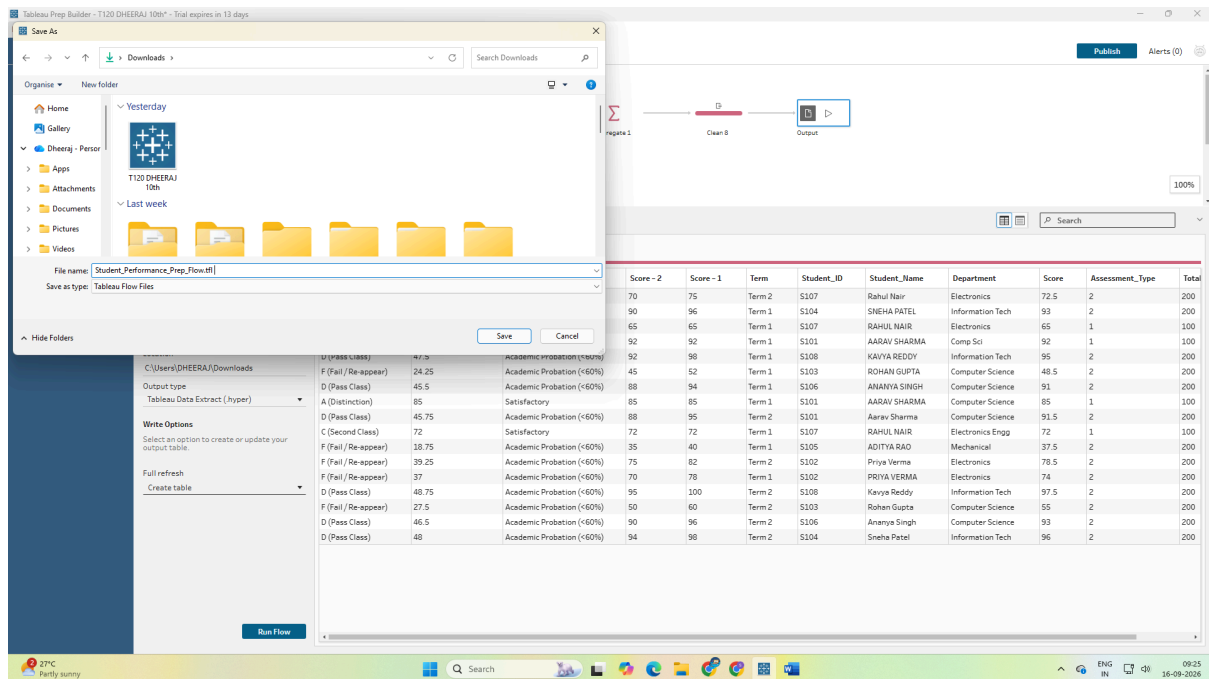
Done

Term	Student_ID	Student_Name	Department	Score	Assessment_Type	Total
Term 2	S107	Rahul Nair	Electronics	72.5	2	200
Term 1	S104	SNEHA PATEL	Information Tech	93	2	200
Term 1	S107	RAHUL NAIR	Electronics	65	1	100
Term 1	S101	AARAV SHARMA	Comp Sci	92	1	100
Term 1	S108	KAVYA REDDY	Information Tech	95	2	200
Term 1	S103	ROHAN GUPTA	Computer Science	48.5	2	200
Term 1	S106	ANANYA SINGH	Computer Science	91	2	200
Term 1	S101	AARAV SHARMA	Computer Science	85	1	100
Term 2	S101	Aarav Sharma	Computer Science	91.5	2	200
Term 1	S107	RAHUL NAIR	Electronics Engg	72	1	100
Term 1	S105	ADITYA RAO	Mechanical	37.5	2	200
Term 2	S102	Priya Verma	Electronics	78.5	2	200
Term 1	S102	PRIYA VERMA	Electronics	74	2	200
Term 2	S108	Kavya Reddy	Information Tech	97.5	2	200
Term 2	S103	Rohan Gupta	Computer Science	55	2	200
Term 2	S106	Ananya Singh	Computer Science	93	2	200
Term 2	S104	Sneha Patel	Information Tech	96	2	200

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Step 4: Save and Automate the Flow File

1. Go to the top menu bar $\rightarrow$ select File $\rightarrow$ Save As....
2. Name your workflow : Student_Performance_Prep_Flow.tfl

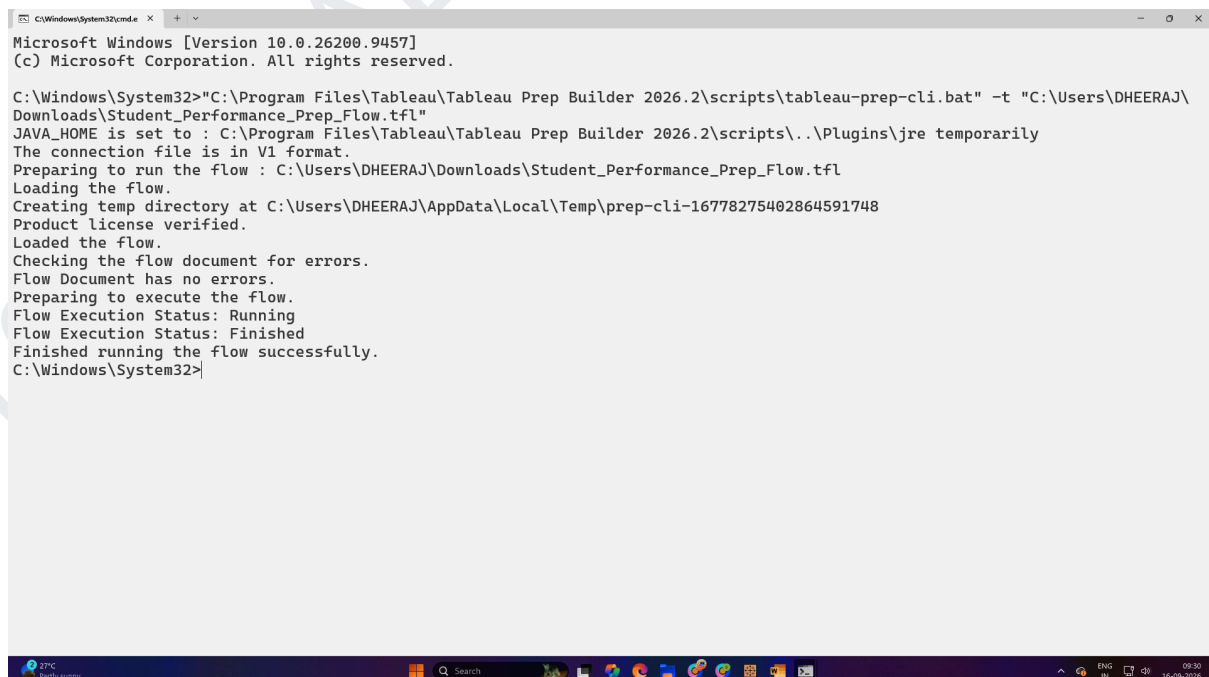


3. (Automation Options in Production):

o Command Line Execution: You can automate this flow on Windows/Mac using the Tableau Prep Command Line Interface (CLI):

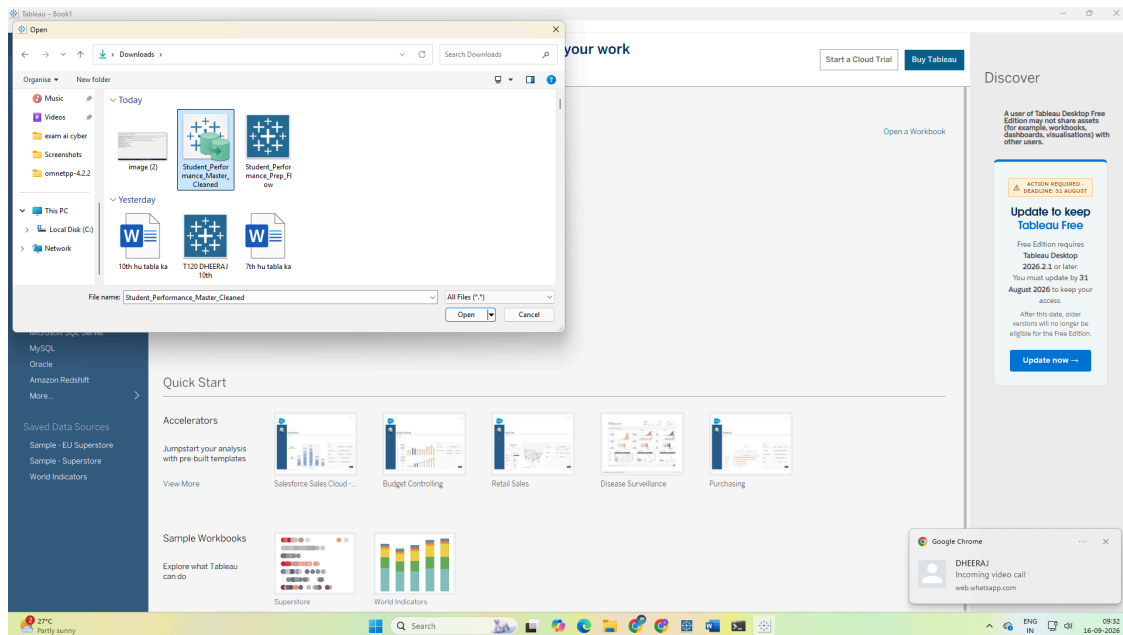
Bash : `tableau-prep-cli -t Student_Performance_Prep_Flow.tfl`

o Tableau Prep Conductor: If connected to Tableau Cloud/Server, you can publish the flow and set a scheduled refresh (e.g., every Monday at 6:00 AM) to automatically run whenever new semester files are added.



Step 5: Verify the Final Master Dataset in Tableau Desktop

1. Open Tableau Desktop.
2. Connect to the exported Student_Performance_Master_Cleaned.csv (or .hyper).



3.

Build the self-performance dashboard using the pre-calculated metrics:

- o Drag Student_Name to Rows.
- o Drag Term to Columns.
- o Drag Term_Percentage to Text.
- o Drag Letter_Grade to Color.
- o Drag Academic_Standing to Tooltip

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